



# Onboard cooperative relative positioning system for Micro-UAV swarm based on UWB/Vision/INS fusion through distributed graph optimization

Chengsong Xiong<sup>a,b,\*</sup>, Wenshuai Lu<sup>a,b</sup>, Hao Xiong<sup>a,b</sup>, Haolun Ding<sup>a,b</sup>, Qingning He<sup>a,b</sup>, Duo Zhao<sup>a,b</sup>, Jiaqi Wan<sup>a,b</sup>, Fei Xing<sup>a,b</sup>, Zheng You<sup>a,b,\*</sup>

<sup>a</sup> Department of Precision Instrument, Tsinghua University, Beijing 100084, China

<sup>b</sup> State Key Laboratory of Precision Measurement Technology and Instruments, Tsinghua University, Beijing 100084, China

## ARTICLE INFO

### Keywords:

micro-UAV swarm  
Relative positioning  
DGO

## ABSTRACT

In recent years, the swarm of micro Unmanned Aerial Vehicles (micro-UAVs) is becoming a research hotspot. As the basis of swarm autonomous flight, traditional relative positioning methods typically rely on external base stations or onboard sensors of high power consumption, the vast majority of which are not feasible on resource-constrained micro-UAVs flying in a complex environment. This article proposes a lightweight relative positioning system implemented on a micro-UAV platform weighing 150 g. We have developed a distributed graph optimization (DGO) scheme fusing the results of the onboard Ultra-Wideband (UWB) module, camera, and inertial sensors. A Delay Extended Kalman Filter (DEKF) is utilized to mitigate communication delays and enhance accuracy. This scheme enables micro-UAVs to collaboratively compute the relative position graph in real-time. Through real-world experiments, our system has achieved a relative positioning accuracy of 0.167 m, exhibiting significant potential for future swarm applications.

## 1. Introduction

In recent years, the field of UAV have attracted increasing research interest. UAVs have great practical value in various applications, including disaster rescue missions [1], environmental monitoring [2], and smart agriculture [3,4], etc. Among the existing various types of UAVs, micro-UAVs weighing a few hundred grams are popular due to their low cost, portability, and maneuverability. However, the mission capability of an individual micro-UAV is severely limited by its hardware resources [5]. Therefore, people usually adopt micro-UAV swarms to accomplish complex tasks such as collaborative monitoring of critical locations [6], regional exploration and mapping [7], and emergency rescue [8], among others. A swarm of micro-UAVs has advantages of higher task efficiency, more payload capacity, and better fault tolerance compared to a single micro-UAV.

The positioning technology serves as the basis for a swarm of UAVs to accomplish tasks cooperatively. However, the majority of current popular positioning methods heavily rely on sensors, references, and ground stations external to the UAV itself. To this day, most researchers have adopted external base stations to obtain the position of UAVs in the ground coordinate system. In outdoor environments, the mainstream

positioning methods involve fusing the estimation results from the inertial navigation system (INS) and global navigation satellite system (GNSS) [9,10]. However, when the GNSS is unavailable in complex electromagnetic environment and indoor scenario [11], each node in the swarm can only obtain the positions of other nodes through INS. Due to the drift of IMU, this method usually leads to drifting and unreliable results over time. In indoor environments, researchers utilize centralized measuring equipment such as motion capture systems [12,13] and UWB base stations [14,15] to achieve high-precision positioning at the centimeter level. The UAV swarms relying on these bulky pre-deployed devices can only fly in the laboratory environment and require substantial hardware resources for centralized communication and computation. A fundamental requirement for an autonomous robotic swarm is the capability to be swiftly and effortlessly deployed in various scenarios, which the previously mentioned systems fail to meet. Hence, their practical value is restricted.

The relative positioning scheme achieved solely through onboard sensors of the UAVs has gained significant research attention in the past decade, owing to the limitations of the aforementioned systems. Among these studies, methods based on visual-inertial odometry (VIO) are most commonly used in estimating the positions of UAVs. One of the pioneer

\* Corresponding authors.

E-mail addresses: [xcs20@mails.tsinghua.edu.cn](mailto:xcs20@mails.tsinghua.edu.cn) (C. Xiong), [yz-dpi@mail.tsinghua.edu.cn](mailto:yz-dpi@mail.tsinghua.edu.cn) (Z. You).

<https://doi.org/10.1016/j.measurement.2024.114897>

Received 31 January 2024; Received in revised form 22 April 2024; Accepted 10 May 2024

Available online 13 May 2024

0263-2241/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

works applied a high-precision VIO system to estimate the pose of each UAV in a swarm [16]. The relative distances between the UAVs were measured by onboard UWB modules. Then a simple optimization problem was formulated, which corrects the drift of VIO in the long term. Other methods that aim to enhance the performance of VIO or employ external sensors to correct VIO drift have also been extensively examined [17,18]. Recently, another type of relative positioning method based on low-cost sensor arrays has also been investigated [19,20]. For instance, in [19], four UWB nodes were mounted on a single robot, thus forming multiple sets of distance measurements between two robots. In addition, the relative orientation between two robots can also be obtained based on this measurement system. Other notable approaches include distance and orientation estimation using a monocular camera [21,22], beamforming of acoustic signals [23], and relative measurements using LiDAR [24], among others.

These methods contribute significantly to achieving autonomous flight of UAV swarm; however, they have limitations in terms of universal applicability. VIO-based methods strongly depend on high-performance computing devices. For instance, in [16], a NVIDIA Xavier CPU with 6 cores and 8 GB memory was mounted on the UAV. About 11 % of the CPU computing power was used for relative positioning, which is several times higher than that used for planning and mapping. For general micro-UAVs, their batteries and sizes are insufficient to support computing devices of this level. Methods based on sensor arrays, as discussed in [19,20], enhance measurement accuracy by deploying multiple sensors of the same type on a single UAV, thereby providing redundant measurement information. These methods rely on the premise that the installation distances between sensors are adequate to estimate relative angles, while the limited size of micro-UAVs poses a challenge in meeting this condition. Other methods like monocular vision [21,22] are not restricted by the size of UAV, since micro cameras are readily available commercially. However, the limited field of view (FOV) of monocular cameras is a significant disadvantage. Consequently, a UAV can only detect a limited number of cooperative targets with a single monocular camera. As the number of UAVs increases, this limitation becomes a fatal problem. The same drawback also occurs with LiDAR-based methods [24]. These challenges impede the widespread adoption of existing relative positioning methods in general multi-UAV systems, especially for low-power, miniaturized, and computationally limited micro-UAVs.

To address these issues, there exists an urgent need to explore new lightweight relative positioning methods that can be fully achieved using the onboard sensors of UAVs. The aim is to effectively utilize the sensing, communication, and computing resources of UAV swarms to enable each UAV to obtain its relative position to other nodes. In this paper, we propose a fast and high-precision relative positioning system designed specifically for micro-UAV swarms. The proposed onboard measurement system utilizes UWB nodes for distributed ranging. Meanwhile, a compressed YOLOv7-tiny network is implemented on an embedded lightweight AI processor for cooperative target detection and relative direction calculation based on monocular vision [25,26]. In addition, the system fuses the INS estimations of the UAV through an extended Kalman filter (EKF) for self-state estimation. We model the relative positioning of UAVs as a graph optimization (GO) problem [27,28]. The results of the aforementioned three types of sensors are fused into a rough relative position estimation. Considering the computing power and memory limitation of the micro-UAV, we propose the DGO scheme, in which the global position graph of the UAV swarm is decomposed into multiple subgraphs. Each UAV parallelly solves a local optimization problem and exchanges the results with other UAVs in the swarm. To further improve accuracy and compensate for delay, we have designed a fusion system based on DEKF [29,30]. Combining the DGO method with DEKF, we acquire smoother and more precise relative positioning results. Numerical simulations indicate that the proposed method can provide relatively accurate positioning results even with limited sensing accuracy. Furthermore, we deployed the proposed

algorithm on a real-world micro-UAV system and conducted experiments to assess its performance. The results show that our algorithm has achieved relative positioning accuracies of 0.167 m at a maximum flight velocity of 0.82 m/s and 0.193 m at a maximum flight velocity of 2.09 m/s respectively, despite the limitations in sensing, computing, and communication resources of the micro-UAV. The proposed method outperforms other methods under the same hardware condition.

The main contributions of this paper are as follows:

- A lightweight onboard measurement system for micro-UAV swarm is proposed. With multiple MEMS sensors fusion and neural networks combined, the system achieves a distance measurement accuracy of 0.08 m and an angle measurement accuracy of  $2.3^\circ$ .
- The collaborative relative positioning method based on DGO and DEKF is developed. Compared to existing high-power and high-computing relative positioning systems, the proposed method can significantly simplify calculations without sacrificing accuracy. This method maintains high measurement precision and demonstrates satisfactory real-time capability on micro-UAV platforms.
- The proposed measurement system and relative positioning method are implemented on a lightweight micro-UAV platform weighing 150 g, developed by our team. The precision and practicality of the proposed method are demonstrated through real-world experiments.

## 2. Measurement system of the UAV swarm

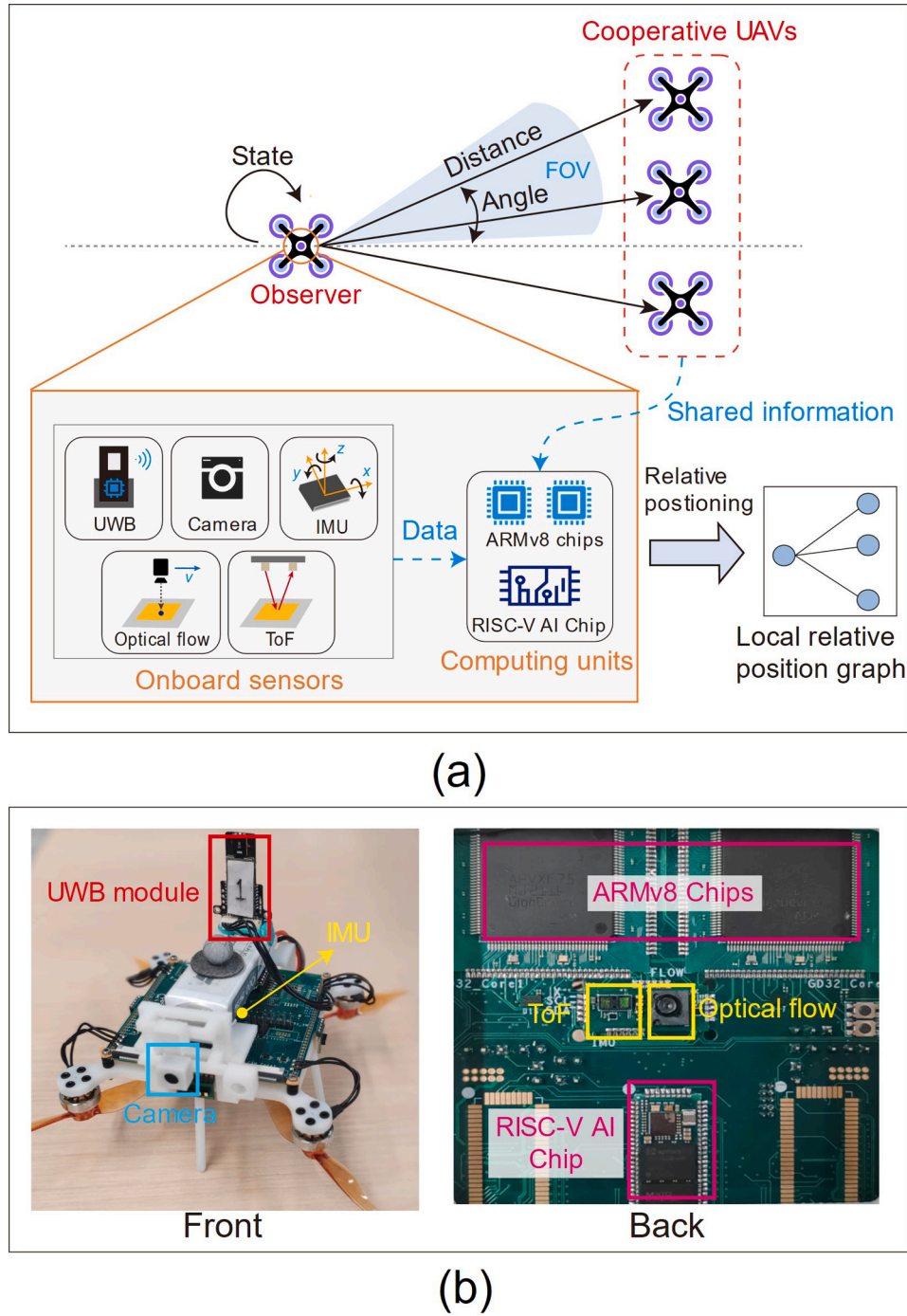
In this section, the measurement system of the UAV swarm and sensor models are described. Typically, the measurement system of a UAV comprises three types of sensors: distance measurement, angle measurement and state estimation sensors. In this study, a UWB module, monocular camera, MEMS inertial measurement unit (IMU), optical flow sensor, and laser time-of-flight (ToF) sensor are mounted on the UAV, as shown in Fig. 1. These MEMS sensors are at millimeter or centimeter level dimensions, to ensure the integrability of the system. The computing units of the UAV consist of two GigaDevice ARMv8 Cortex-M chips and a GreenWaves RISC-V AI processor. The computing units process local sensor data and information received from cooperative UAVs to calculate relative positions.

### 2.1. UWB for relative distance measurement

UWB technology has been widely applied in wireless sensor networks (WSN) [31–33], ground multi-robot systems [34,35] and UAV swarms [14–16,19,36] during the past few years. UWB holds a great advantage of omnidirectionality compared to other range-finding sensors like monocular cameras [21,22] and LiDAR [24]. In some related studies, multiple UWB nodes are mounted on the same robot to calculate relative positions [19,20]. This method is not suitable for our swarm due to the limited size and battery capacity of the micro-UAVs. In this study, we aim to leverage the potential of the distributed distance measuring mode of UWB to achieve an anchor-free UAV swarm. A  $41 \times 27 \times 7$  mm LinkTrack P-BS UWB module produced by NOOLOOP is mounted on each UAV to estimate its distances to other UAVs, without pre-deployed anchors or ground base stations. The distance measurement noise can be modelled as Gaussian distribution. Considering a swarm consisting of  $N$  UAVs, the UWB measurement process between UAV (i) and UAV (j) can be formulated as

$$d_{k,(i,j)}^{UWB} = h^{UWB}(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}) + N(0, \Sigma_d^2) \quad (1)$$

where  $d_{k,(i,j)}^{UWB}$  is the relative distance measurement result of UWB between the two UAVs at epoch  $k$ ,  $h^{UWB}(\cdot, \cdot)$  stands for the observation function.  $\mathbf{p}_{k,(i)}$  and  $\mathbf{p}_{k,(j)}$  are the positions of UAVs, and  $\Sigma_d^2$  is the variance of Gaussian white noise. In this section, we establish sensor models and conduct pre-experiments to estimate the noise level. The measurement error determines the weights of sensors in the proposed fusion system,



**Fig. 1.** Measurement system of UAV. (a) The diagram of relative position measurement. (b) Photographs of the micro-UAV platform and sensors.

which will be detailed in section 3.

## 2.2. Monocular camera for relative angle measurement

A variety types of direction-finding sensors have been implemented on multi-UAV systems, including acoustic beamforming [23], wireless beamforming [37], and vision-based methods [21,22]. Beamforming-based methods require multiple sensors, and the precision of results is limited by the size of the sensor array. For vision-based methods, relative angle measurement can be achieved through cooperative target recognition with a single resolution-limited micro monocular camera, based on the lightweight neural network deployed on micro-UAV. We have designed a relative positioning scheme that requires only a few

pairs of relative angle measurements, in order to overcome the limitation of the camera's FOV. In this study, an *OmniVision* OV5647 camera with a pixel resolution set to  $320 \times 240$  and a FOV of  $72.9^\circ$  is applied. To distinguish cooperative UAVs from the background environment without pre-deployed optical markers, we trained a lightweight YOLOv7-tiny neural network of 1.5 M parameters, 828 M FLOPS and 99.2 % precision [25,26]. The network was deployed on the RISC-V AI processor, which runs at 15 fps and supports  $320 \times 240$  pixels image input, as shown in Fig. 2.

After the cooperative target is recognized and marked in the two-dimensional pixel plane, the relative angle can be estimated based on the triangulation principle shown in Fig. 3. In the camera coordinate system, we assume the target point is located at  $(X_c, Y_c, Z_c)$ , and its



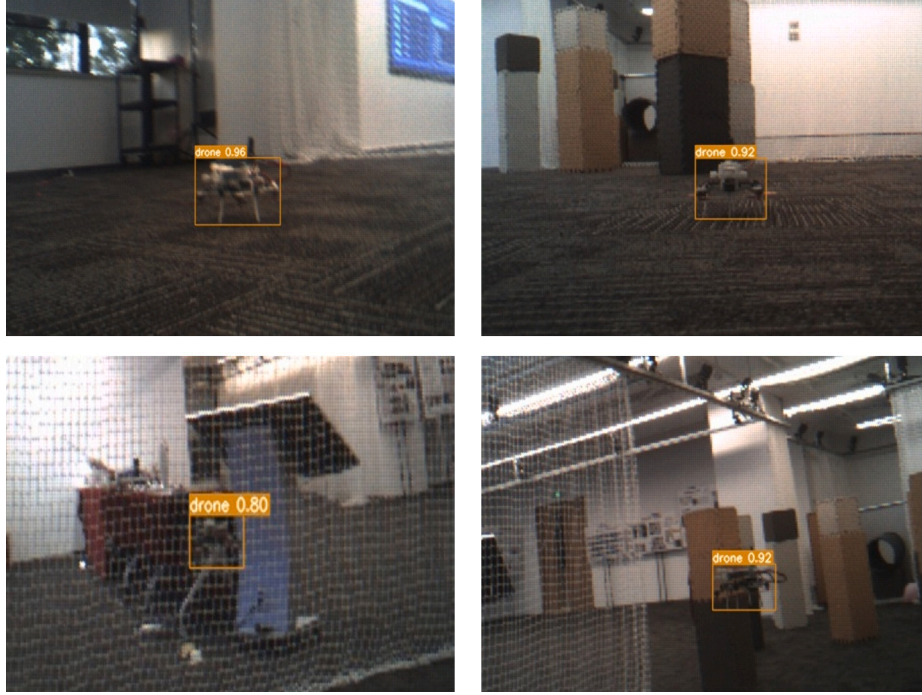


Fig. 2. Cooperative UAV targets detected by YOLOv7-tiny neural network.

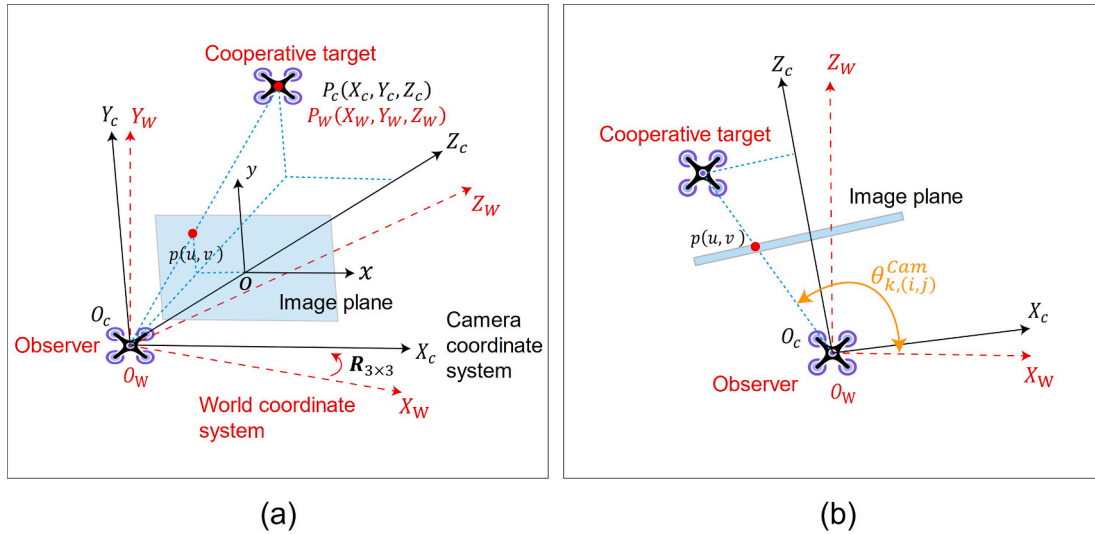


Fig. 3. The diagram of relative angle measurement. (a) Diagram in the 3D space. (b) Diagram on the horizontal plane.

projection on the image plane is  $(u, v)$ . We define the world coordinate system as follows: the origin is fixed to the UAV, with the X-axis pointing east, the Z-axis pointing north, and the Y-axis pointing vertically upward. In the world coordinate system, the target point is located at  $(X_w, Y_w, Z_w)$ . According to the principle of monocular camera imaging, we have

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} [R_{3 \times 3} \quad T_{3 \times 1}] \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix} \quad (2)$$

where  $f_x$  and  $f_y$  represent the focal length in the x and y directions respectively.  $u_0$  and  $v_0$  are pixel coordinates corresponding to the optical center of the camera.  $R_{3 \times 3}$  is the rotation matrix between the world

coordinate system and camera coordinate system, which can be directly calculated from IMU output. The translation vector  $T_{3 \times 1}$  can be approximated to  $0_{3 \times 1}$ , since the distance between the UAV geometric center and the camera is negligibly small. Then equation (2) can be simplified as

$$Z_c \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} R_{3 \times 3} \begin{bmatrix} X_w \\ Y_w \\ Z_w \end{bmatrix} \quad (3)$$

Although  $Z_c$  is unknown, when  $(u, v)$  is obtained from the YOLOv7 network, we can calculate the direction vector from the observer to the target UAV:



$$\Theta = \frac{1}{Z_c} \begin{bmatrix} X_W \\ Y_W \\ Z_W \end{bmatrix} \quad (4)$$

Then, the relative angle between UAVs on the horizontal plane can be straightforwardly calculated. It is important to address distortion issues that may arise during the imaging process. Therefore, we also implement a calibration step to ensure accurate and reliable results.

The measurement process of relative angle can be written as

$$\theta_{k,(i,j)}^{Cam} = h^{Cam}(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}) + N(0, \Sigma_{\theta}^2) \quad (5)$$

where  $\theta_{k,(i,j)}^{Cam}$  stands for the relative angle measurement result between UAV (i) and UAV (j) at epoch  $k$  on the horizontal plane,  $h^{Cam}(*, *)$  is the observation function, and  $\Sigma_{\theta}^2$  is the variance of Gaussian white noise.

### 2.3. INS for self-state estimation

Generally, the INS of a UAV consist of a series of sensors, including an accelerometer, gyroscope, magnetometer, barometer, and camera, among others [38]. In the GNSS-denied environment, VIO is widely applied as a self-state estimation approach, which fuses the estimations of camera and inertial sensors [16–18,39]. However, the deployment of VIO on micro-UAVs is limited by its high power consumption and computational requirements. In this paper, we propose an INS system composed of a MEMS IMU, optical flow sensor, and ToF sensor to accommodate the hardware resources available on micro-UAVs.

The measurement process of IMU can be simplified as

$$\begin{bmatrix} \mathbf{a}_{k,(i)}^{IMU} \\ \boldsymbol{\omega}_{k,(i)}^{IMU} \end{bmatrix}^T = h^{IMU}(\mathbf{X}_{k,(i)}) + \mathbf{N}^{IMU} + \mathbf{B}^{IMU} \quad (6)$$

in which,  $\mathbf{a}_{k,(i)}^{IMU}$  is the three-axis acceleration vector of UAV,  $\boldsymbol{\omega}_{k,(i)}^{IMU}$  is the three-axis angular velocity vector, and  $h^{IMU}(*, *)$  is the observation function of the IMU. The true state vector of the UAV denoted as  $\mathbf{X}_{k,(i)}$  is used as the input of  $h^{IMU}(*, *)$ .  $\mathbf{N}^{IMU}$  and  $\mathbf{B}^{IMU}$  represent the noise matrix and bias matrix of the IMU respectively.

The optical flow sensor is widely applied in UAV systems for velocity measurement as a supplement to IMU [40,41]. The optical flow measurement model is given by

$$\mathbf{V}_{k,(i)}^{flow} = h^{flow}(\mathbf{X}_{k,(i)}) + \mathbf{N}^{flow} \quad (7)$$

where  $\mathbf{V}_{k,(i)}^{flow}$  is the velocity of the UAV (i) obtained from the optical flow sensor,  $h^{flow}(*, *)$  is the observation function, and  $\mathbf{N}^{flow}$  is the noise matrix of the optical flow sensor.

To estimate the flight height accurately, a ToF sensor is installed on the bottom of the UAV. By calculating the flying time of laser, the UAV obtains cm-level height estimations. The measurement model of the ToF sensor can be written as

$$p_{z,k,(i)}^{ToF} = h^{ToF}(\mathbf{X}_{k,(i)}) + N(0, \Sigma_z^2) \quad (8)$$

where  $p_{z,k,(i)}^{ToF}$  represents the height of the UAV measured by ToF.

For further performance improvement of INS, EKF is utilized to fuse the output of the aforementioned MEMS sensors. To simplify the expressions, the true state vector of the UAV is written as

$$\mathbf{x} = [\mathbf{p}, \mathbf{v}, \mathbf{q}, \mathbf{a}_b, \boldsymbol{\omega}_b, \mathbf{g}]^T \in \mathbb{R}_{18 \times 1} \quad (9)$$

in which,  $\mathbf{p} = [p_x, p_y, p_z]^T$  is the 3-axis position of the UAV. Since the INS calculates displacement by integration, the starting point is taken as reference.  $\mathbf{v} = [v_x, v_y, v_z]^T$  is the 3-axis velocity, and  $\mathbf{q} = [q_x, q_y, q_z, q_w]^T$  is the pose in quaternion form.  $\mathbf{a}_b$  and  $\boldsymbol{\omega}_b$  represent the bias of acceleration and angular velocity respectively.  $\mathbf{g}$  is the gravitational acceleration. The measurements of acceleration and angular velocity are

described as

$$\mathbf{u}_m = [\mathbf{a}_m, \boldsymbol{\omega}_m]^T \in \mathbb{R}_{6 \times 1} \quad (10)$$

For UAV (i) at epoch  $k$ ,  $\mathbf{u}_m$  is obtained from IMU output. After all state vectors are initialized, the state at the next timestep can be predicted using IMU measurements. Assuming  $\Delta t$  is the time interval,  $\mathbf{R} = \mathbf{R}\{\mathbf{q}\}$  is the rotation matrix generated from  $\mathbf{q}$ . Then the prediction process of the state of discrete-time is formulated as

$$\begin{aligned} \mathbf{p} &\leftarrow \mathbf{p} + \mathbf{v}\Delta t + \frac{1}{2}(\mathbf{R}(\mathbf{a}_m - \mathbf{a}_b) + \mathbf{g})\Delta t^2 \\ \mathbf{v} &\leftarrow \mathbf{v} + (\mathbf{R}(\mathbf{a}_m - \mathbf{a}_b) + \mathbf{g})\Delta t \\ \mathbf{q} &\leftarrow \mathbf{q} \otimes \mathbf{q}\{(\boldsymbol{\omega}_m - \boldsymbol{\omega}_b)\Delta t\} \end{aligned} \quad (11)$$

Equation (11) can be written as

$$\mathbf{x} \leftarrow f(\mathbf{x}, \mathbf{u}_m) = \mathbf{A}_x \mathbf{x} \quad (12)$$

where  $f(\mathbf{x}, \mathbf{u}_m)$  is the state transition function, and  $\mathbf{A}_x$  is the Jacobian matrix generated from  $f(\mathbf{x}, \mathbf{u}_m)$ . The error state covariance matrix  $\mathbf{P}$  is updated according to the following equation:

$$\mathbf{P} \leftarrow \mathbf{A}_x \mathbf{P} \mathbf{A}_x^T + \mathbf{Q} \quad (13)$$

where  $\mathbf{Q}$  is the process noise matrix. The estimation results of optical flow and ToF sensors are used as observation vectors for the INS. The Kalman gain  $\mathbf{K}$ , state vector  $\mathbf{x}$ , and error state covariance matrix  $\mathbf{P}$  are updated:

$$\begin{aligned} \mathbf{K} &= \mathbf{P} \mathbf{H}^T (\mathbf{H} \mathbf{P} \mathbf{H}^T + \mathbf{V})^{-1} \\ \mathbf{x} &\leftarrow \mathbf{K}(\mathbf{z} - h(\mathbf{x})) \\ \mathbf{P} &\leftarrow (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P} \end{aligned} \quad (14)$$

in which,  $\mathbf{H}$ ,  $\mathbf{z}$ , and  $h(\mathbf{x})$  represent the observation matrix, observation value, and observation function respectively.

Utilizing EKF, we obtain smoother and more accurate position estimations from the INS. However, it should be noted that the observation vector obtained from the optical flow sensor still accumulates errors over prolonged periods of flight. To address this, the introduction of relative measurements becomes crucial for correcting these drifts.

### 2.4. Evaluation of measurement system accuracy

We have conducted research on the accuracies achieved by individual measurement systems under dynamic conditions. The data of sensors was obtained during the flight process of UAV. The original data and error curves are presented in Fig. 4 and Supplementary Materials, and the detailed parameters are listed in Table 1. The relative measurement accuracy of UWB and camera reaches 0.08 m and 2.3°, respectively. It should be noted that a single UWB system cannot fulfill the purpose of relative positioning, since this work applies an anchor-free UWB network, in order to explore methods for the autonomous flight of UAV swarms independent of ground stations. Without prior knowledge of UWB node positions, the network only provides distributed distance measurement information, which is insufficient for the three-dimensional positioning required for swarm flight. Combining UWB and camera measurements, relative positions can be roughly estimated, however, the measurement range of this method is limited by the camera's FOV. For an individual UAV within a swarm, obtaining the positions of other cooperative targets that are not within its camera view is crucial. The self-state estimation system directly provides 3-axis relative positions; however, its error accumulates over time and the results gradually become unreliable, as shown in Fig. 4. Due to numerous limitations of individual measurement systems, it is necessary to develop a fusion method that fully utilizes measurement information to provide three-dimensional relative positions, while minimizing computing power and communication burden.

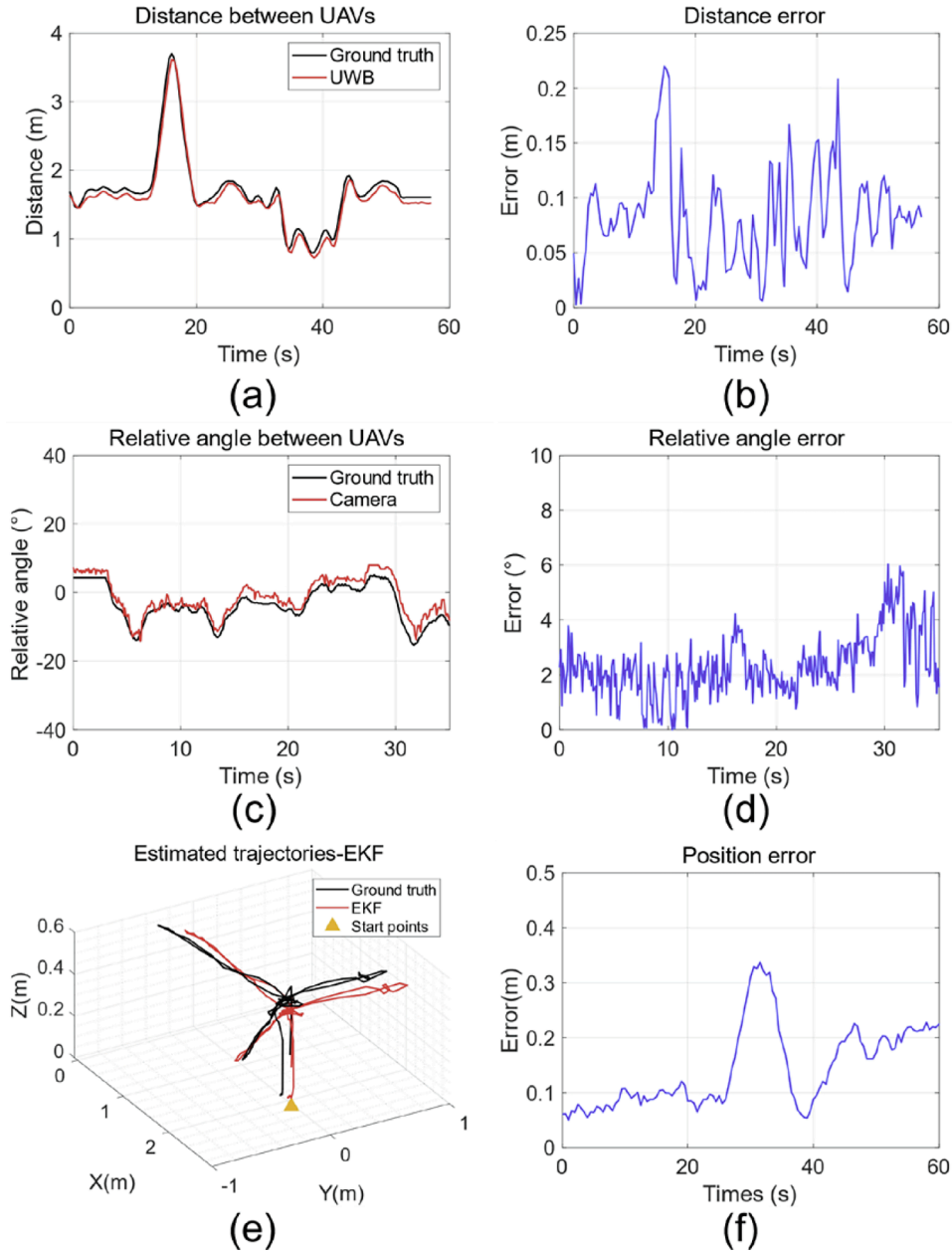


Fig. 4. Measurements and error curves of individual measurement systems.

Table 1

Parameters of onboard sensors.

| Sensor        | Update rate | Error                                                                                         |
|---------------|-------------|-----------------------------------------------------------------------------------------------|
| UWB           | 25 Hz       | 0.08 m                                                                                        |
| Camera        | 10 Hz       | 2.3°                                                                                          |
| Accelerometer | 1000 Hz     | Noise density: $70 \mu\text{g}/\sqrt{\text{Hz}}$<br>Bias stability: 0.1 mg                    |
| Gyroscope     | 1000 Hz     | Noise density: $0.0028^\circ/\text{s}/\sqrt{\text{Hz}}$<br>Bias stability: $2^\circ/\text{h}$ |
| Optical flow  | 10 Hz       | 10 %                                                                                          |
| ToF           | 10 Hz       | 0.05 m                                                                                        |

### 3. Relative positioning algorithm for UAV swarm

In this section, the detailed mechanism of the relative positioning system is introduced, as shown in Fig. 5. The system consists of two major parts: firstly, optimizing the positions of the UAVs using relative

measurements and state estimations based on DGO; secondly, employing the designed DEKF to fuse measurements and correct state errors caused by time delay, thereby enhancing positioning accuracy through iterative processes. Each UAV receives optimization results from other UAVs through low-frequency communication and then performs the optimization process on a local solver. The DEKF system is implemented on UAVs within a distributed framework to ensure that each UAV only needs to correct its local relative position estimations. This approach effectively reduces computational complexity.

#### 3.1. Pre-processing of relative measurements

In real UAV flight scenes, relative measurement sensors such as UWB and camera encounter various challenges and disturbances. Therefore, removing outliers is necessary in the pre-processing system. For UWB ranging results, we utilized the Z-score method to distinguish the outliers from the rest [42]. We maintain a queue of UWB ranging results from epoch  $k_1$  to  $k_2$  ( $Q_{k_1 \rightarrow k_2, (i,j)}^{UWB}$ ) while sampling sensor data:

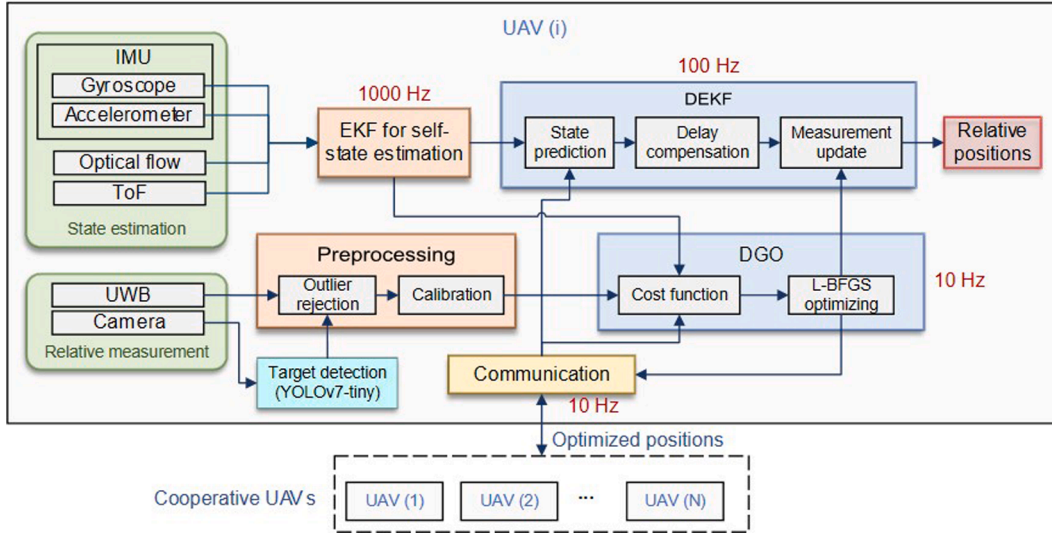


Fig. 5. Diagram of the relative positioning system.

$$\mathbf{Q}_{k_1 \rightarrow k_2, (ij)}^{UWB} = \{d_{k_1, (ij)}^{UWB}, d_{k_1+1, (ij)}^{UWB}, \dots, d_{k_2, (ij)}^{UWB}\} \quad (15)$$

For a sample point  $d_{k, (ij)}^{UWB}$  in the queue, the Z-score is calculated as

$$z(d_{k, (ij)}^{UWB}) = \frac{d_{k, (ij)}^{UWB} - \mu(\mathbf{Q}_{k_1 \rightarrow k_2, (ij)}^{UWB})}{\sigma(\mathbf{Q}_{k_1 \rightarrow k_2, (ij)}^{UWB})} \quad (16)$$

where  $\mu(\cdot)$  and  $\sigma(\cdot)$  represent the average function and standard deviation function respectively. Since the noise of relative distance and angle follows a normal distribution, according to the three-sigma rule, values that are three standard deviations away from the mean have a very low probability of occurrence, and therefore can be considered as outliers. The outlier criterion of UWB ranging results is defined as

$$status = \begin{cases} usage, & \text{if } |z(d_{k, (ij)}^{UWB})| < 3 \\ removal, & \text{else} \end{cases} \quad (17)$$

Similarly, the Z-score can also be applied to judge the quality of the estimated relative angle. The neural network offers an additional output of confidence level ( $cl$ ). Then, we obtain the outlier criterion of relative angle output:

$$status = \begin{cases} usage, & \text{if } |z(\theta_{k, (ij)}^{Cam})| < 3 \text{ and } cl > \varepsilon_{cl} \\ removal, & \text{else} \end{cases} \quad (18)$$

where  $\varepsilon_{cl}$  is the user-defined lower bound of confidence level. Note that when  $cl < \varepsilon_{cl}$ , the network fails to distinguish the UAV from other objects in the camera view and gives wrong recognition results. The measured relative angle is not available under such conditions. To obtain smoother relative measurements, a mean filter is applied to the measured distances and angles so that the Gaussian noise can be suppressed to a certain extent.

Another necessary step of pre-processing is ID matching. Conventionally, without specifically designed optical markers mounted on UAVs [43,44], the UAV is unable to determine the ID of the detected target relying on camera detection results. However, combining the previously known INS results which roughly estimates the relative angle between UAVs, the process of ID matching can be written as

$$j = \underset{j}{\text{find}} \left( \min \left( \left| \angle(\mathbf{p}_{k, (j)}^{INS}, \mathbf{p}_{k, (i)}^{INS}) - \theta_{k, (ij)}^{Cam} \right| \right) \right) \quad (19)$$

where  $i$  represents the ID of the source UAV, and  $j$  is the ID of the

target. The function  $\angle(\cdot, \cdot)$  calculates the relative angle between two given positions.  $\mathbf{p}_{k, (i)}^{INS}$  and  $\mathbf{p}_{k, (j)}^{INS}$  represent the positions of the source UAV and target UAV estimated by INS, respectively.

### 3.2. Distributed graph optimization for relative positioning

In this section, the relative positioning scheme of the UAV swarm based on DGO is introduced. While centralized optimization may achieve higher accuracy, it also results in increased communication burden and computational overhead that is proportional to the number of nodes. For our resource-limited micro-UAV platform, paying such costs for minor improvements in accuracy is not worthwhile. The diagram of the proposed DGO method is shown in Fig. 6. At each time step, each UAV obtains position fusion results of other UAVs through communication and solves an optimization problem with local measurement constraints. The distance constraint angle constraint, relative position constraint are constructed using UWB and camera results. The odometer constraint optimizes the UAV's position based on the displacement between the previous algorithm step (epoch  $k-1$ ) and the current step (epoch  $k$ ). In this scheme, we take the omnidirectionality of the sensors into consideration, especially the camera. It is challenging for the camera with  $72.9^\circ$  FOV to simultaneously capture all the other UAVs; however, our algorithm adapts well to such scenarios. As shown in Fig. 6, if UAV (i) can see UAV (j) in the camera view, a relative angle error cost function and a relative position function are formulated. These constraints contribute to improving the relative measurement precision between the two UAVs. Otherwise, for the UAVs not in the camera view, the relative position can be optimized using the distance error cost function and state estimation, since the UWB module is omnidirectional. This approach maintains the system's stability, ensuring that the UAVs can be accurately positioned even when they are out of the camera's view. The optimized position is then shared with other UAVs in the next time step. Eventually, the relative position estimation of the entire swarm will converge to a relatively steady state. The principle of the scheme is detailed as follows.

We consider a swarm containing  $N$  UAVs. The true position of UAV (i) at epoch  $k$  is defined as  $\mathbf{p}_{k, (i)} = [x_{k, (i)}, y_{k, (i)}, z_{k, (i)}]^T$ . Then the true positions of the UAV swarm are given by

$$\mathbf{P}_k = \{\mathbf{p}_{k, (1)}, \mathbf{p}_{k, (2)}, \dots, \mathbf{p}_{k, (N)}\} \quad (20)$$

For UAV (i), the relative distances to other UAVs at epoch  $k$  are obtained from the UWB sensor:



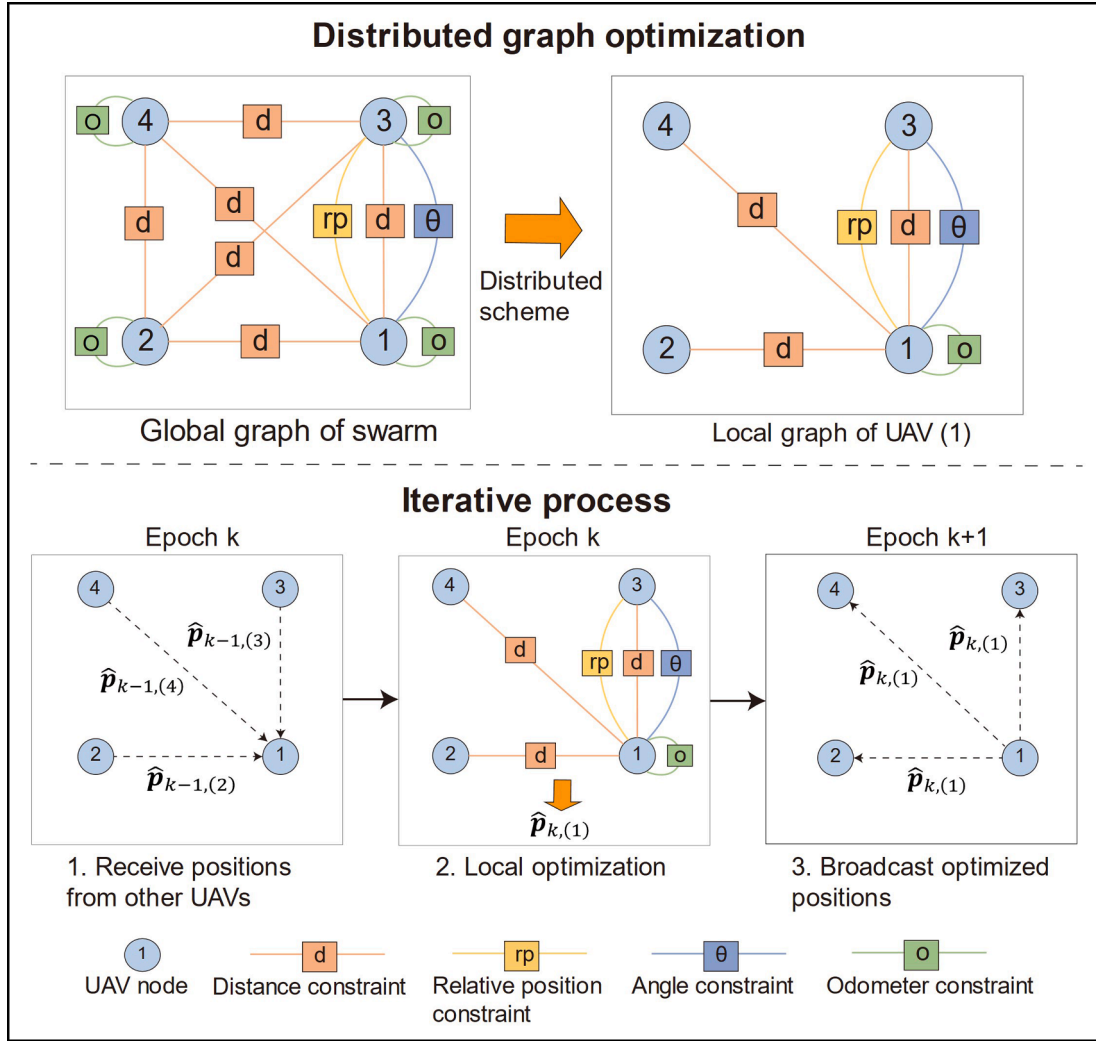


Fig. 6. Diagram of the distributed graph optimization.

$$\mathbf{D}_{k,(i)}^{UWB} = \{d_{k,(ij)}^{UWB}\}, j \in \mathcal{C}_d \quad (21)$$

where  $d_{k,(ij)}^{UWB}$  denotes the distance between UAV (i) and UAV (j) measured by UAV (i). The set  $\mathcal{C}_d$  stores number  $j$ , if UAV (j) can be observed by the UWB sensor of UAV (i). In the proposed distributed optimization architecture, each UAV only utilizes its local sensor information for computation rather than the sensor data of all UAVs, in order to reduce communication burden. At epoch  $k$ , each UAV broadcasts its own position estimated at epoch  $k-1$ :

$$\hat{\mathbf{P}}_{k-1} = \{\hat{\mathbf{p}}_{k-1,(1)}, \hat{\mathbf{p}}_{k-1,(2)}, \dots, \hat{\mathbf{p}}_{k-1,(N)}\} \quad (22)$$

To compensate for the displacement of UAVs between epoch  $k-1$  and  $k$ , the velocity is also shared with other UAVs. In our system, only a few bytes are required to broadcast this information. Then, UAV (i) acquires the positions of other UAVs after compensation:

$$\mathbf{P}_k^C = \{\mathbf{p}_{k,(1)}^C, \mathbf{p}_{k,(2)}^C, \dots, \mathbf{p}_{k,(N)}^C\} \quad (23)$$

$$\mathbf{p}_{k,(j)}^C = \hat{\mathbf{p}}_{k-1,(j)} + \mathbf{v}_{k-1,(j)}^{\text{flow}} \Delta t, i \neq j \quad (24)$$

where  $\mathbf{p}_{k,(i)}^C$  is the compensated position of UAV (i), and  $\Delta t$  is the time interval between two epochs. The distance error function of UAV (i) can be expressed as

$$f^d(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}) = d_{k,(ij)}^{UWB} - \|\mathbf{p}_{k,(i)} - \mathbf{p}_{k,(j)}\|_2 \quad (25)$$

Then, we define the error penalty of distance measurement at epoch  $k$  as  $J_{k,(i)}^d$ , which is written as

$$J_{k,(i)}^d = \sum_{j \in \mathcal{C}_d} \left\| d_{k,(ij)}^{UWB} - \|\mathbf{p}_{k,(i)} - \mathbf{p}_{k,(j)}^C\|_2 \right\|_{\Sigma_d}^2 \quad (26)$$

where  $\|(\cdot)\|_{\Sigma}^2 = (\cdot)^T \Sigma^{-1} (\cdot)$  is the Mahalanobis norm of matrix  $(\cdot)$ .

Similarly, the cost function of relative angles can also be given. It should be noticed that the angle information matrix is sparser than the distance owing to the limited FOV of the camera. Even so, the swarm topology shape can be determined by a few pairs of relative angle measurements and sufficient distance constraints. The relative angle measurements are given by

$$\Theta_{k,(i)}^{\text{Cam}} = \{\theta_{k,(ij)}^{\text{Cam}}\}, j \in \mathcal{C}_\theta \quad (27)$$

where  $\theta_{k,(ij)}^{\text{Cam}}$  is the measured relative angle between UAV (i) and UAV (j) at epoch  $k$ . The set  $\mathcal{C}_\theta$  stores number  $j$ , if UAV (j) can be observed by the camera of UAV (i). The relative angle error function can be given by

$$f^\theta(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}) = \theta_{k,(ij)}^{\text{Cam}} - \angle(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}) \quad (28)$$

The angle measurement error penalty is defined as

$$J_{k,(i)}^\theta = \sum_{j \in \mathcal{C}_\theta} \left\| \theta_{k,(ij)}^{Cam} - \angle(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}^C) \right\|_{\Sigma_\theta}^2 \quad (29)$$

As supplementary, once we simultaneously obtain the relative distance and relative angle between two UAVs, the relative position can be directly calculated from the measurement results. An additional constraint is designed to further improve positioning accuracy. The error function is written as

$$f^{rx}(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}) = d_{k,(ij)}^{UWB} \cos(\theta_{k,(ij)}^{Cam}) - |p_{xk,(j)} - p_{xk,(i)}| \quad (30)$$

$$f^{ry}(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}) = d_{k,(ij)}^{UWB} \sin(\theta_{k,(ij)}^{Cam}) - |p_{yk,(j)} - p_{yk,(i)}| \quad (31)$$

Note that  $(p_{xk,(j)} - p_{xk,(i)})$  represents the relative X-axis coordinate between UAV (i) and (j), and  $(p_{yk,(j)} - p_{yk,(i)})$  is the relative Y-axis coordinate. The relative position measurement error penalty generated from UWB and camera measurements can be defined as

$$J_{k,(i)}^p = \sum_{j \in \mathcal{C}_d \cap \mathcal{C}_\theta} \left\| f^{rx}(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}^C) \right\|_{\Sigma_{rx}}^2 + \left\| f^{ry}(\mathbf{p}_{k,(i)}, \mathbf{p}_{k,(j)}^C) \right\|_{\Sigma_{ry}}^2 \quad (32)$$

To fully utilize the INS information, the displacement of the UAV (i) from epoch  $k-1$  to epoch  $k$  ( $\Delta \mathbf{p}_{k-1 \rightarrow k,(i)}^{INS}$ ) is calculated. Although the INS positioning error will accumulate over time, the displacement estimation in a short period is relatively reliable. The optimized position of UAV (i) at epoch  $k-1$ , which is denoted as  $\hat{\mathbf{p}}_{k-1,(i)}$ , is known previously through executing the relative positioning algorithm in this subsection iteratively. The odometer error penalty  $J_{k,(i)}^o$  and can then be given as

$$J_{k,(i)}^o = \left\| \Delta \mathbf{p}_{k-1 \rightarrow k,(i)}^{INS} - (\mathbf{p}_{k,(i)} - \mathbf{p}_{k-1,(i)}) \right\|_{\Sigma_o}^2 \quad (33)$$

Utilizing the penalties above, the cost function of the relative position optimizing problem can be constructed. The position of UAV (i) in the swarm at epoch  $k$  is optimized as:

$$\hat{\mathbf{p}}_{k,(i)} = \underset{\mathbf{p}_{k,(i)}}{\operatorname{argmin}} (J_{k,(i)}^d + J_{k,(i)}^\theta + J_{k,(i)}^p + J_{k,(i)}^o) \quad (34)$$

The sliding window optimization mechanism is applied to improve positioning accuracy. Suppose  $\hat{\mathbf{p}}_{k_1 \rightarrow k_2,(i)}$  represents the estimated positions of UAV (i) from epoch  $k_1$  to  $k_2$ , we have

$$\hat{\mathbf{p}}_{k_1 \rightarrow k_2,(i)} = \underset{\mathbf{p}_{k_1 \rightarrow k_2,(i)}}{\operatorname{argmin}} \left( \sum_{k=k_1}^{k_2} (J_{k,(i)}^d + J_{k,(i)}^\theta + J_{k,(i)}^p + J_{k,(i)}^o) \right) \quad (35)$$

where  $\hat{\mathbf{p}}_{k_1 \rightarrow k_2,(i)} = \{\hat{\mathbf{p}}_{k_1,(i)}, \hat{\mathbf{p}}_{k_1+1,(i)}, \dots, \hat{\mathbf{p}}_{k_2,(i)}\}$ . Then, the relative position between UAV (i) and UAV (j), which is denoted as  $\hat{\mathbf{p}}_{k,(ij)}^R$ , can be obtained:

$$\hat{\mathbf{p}}_{k,(ij)}^R = \mathbf{p}_{k,(j)}^C - \hat{\mathbf{p}}_{k,(i)} \quad (36)$$

The DGO algorithm is summarized as follows:

**Algorithm 1.** Distributed relative position optimization

**Input:** Relative distance measurements:  $\mathbf{D}_{k,(i)}^{UWB}$  Relative angle measurements:  $\Theta_{k,(i)}^{Cam}$  Optimized positions at the last epoch:  $\hat{\mathbf{p}}_{k-1}$  **Output:** Estimated relative positions  $\hat{\mathbf{p}}_{k,(ij)}^R$

1. Initialize system
2. **for**  $i = 1$  to  $N$
3. UAV (i) obtains  $\mathbf{D}_{k,(i)}^{UWB}$ ,  $\Theta_{k,(i)}^{Cam}$  from sensors
4. UAV (j) receives  $\hat{\mathbf{p}}_{k-1}$  from other UAVs through communication
5. Calculate  $J_{k,(i)}^d$ ,  $J_{k,(i)}^\theta$ ,  $J_{k,(i)}^p$  and  $J_{k,(i)}^o$  using (26) (29) (32) (33)
6. Solve the optimization problem of (36) with on-board L-BFGS architecture and obtain  $\hat{\mathbf{p}}_{k_1 \rightarrow k_2,(i)}$
7. Calculate  $\hat{\mathbf{p}}_{k,(ij)}^R$
8. **end for**

Using this distributed sliding window optimization scheme, the local optimization results of each UAV converge to the global optimal solution gradually. Since the computing burden does not increase rapidly with the number of nodes, this scheme can support a swarm with multiple UAVs for real-time relative positioning.

### 3.3. Delay extended Kalman filter

In this section, we present a summary of the designed DEKF scheme and introduce the time delay compensation strategy. The DGO scheme outlined above requires low computing resources, however, each UAV only estimates the position of itself. The positions of other UAVs in the swarm need to be obtained through communication. In our system, the UWB module is also used for communication except for relative distance measurement. Each UAV broadcasts its local estimated position at a low frequency of 10 Hz, when other UAVs receive the position information, a delay of 100 ms occurs. This limited frequency is caused by the constrained processing capabilities of the MCU. To address this issue, we utilize DEKF for delay compensation, enhancing updating rate and improving positioning accuracy.

Considering UAV (i) in the swarm, at epoch  $k$ , its local position  $\hat{\mathbf{p}}_{k,(i)}$  has been optimized, and the positions of other UAVs at epoch  $k-1$  are obtained through communication. Each UAV initializes a series of DEKF filters locally, taking the estimated relative positions as observed values. The system model can be written as

$$\begin{cases} \mathbf{X}_{k+1}^R = \mathbf{A}_k \mathbf{X}_k^R + \mathbf{W}_k \\ \mathbf{Z}_k^R = \mathbf{H}_k \mathbf{X}_k^R + \mathbf{V}_k \end{cases} \quad (37)$$

where  $\mathbf{X}_k^R$  is the relative state between two UAVs at epoch  $k$ :

$$\mathbf{X}_k^R = \begin{bmatrix} \mathbf{p}_{k,(ij)}^R \\ \mathbf{v}_{k,(ij)}^R \end{bmatrix} = \begin{bmatrix} \mathbf{p}_{k,(j)} - \mathbf{p}_{k,(i)} \\ \mathbf{v}_{k,(j)} - \mathbf{v}_{k,(i)} \end{bmatrix} \quad (38)$$

and  $\mathbf{Z}_k^R$  is the observation value.

Suppose the time delay occurs at epoch  $s$ , that is, the observation value at epoch  $s$   $\mathbf{Z}_s^R$  is not obtained until epoch  $k$ . There exist  $N_f$  filtering periods between  $s$  and  $k$ , as shown in Fig. 7. The process of DEKF is described as follows:

The prediction process is

$$\hat{\mathbf{X}}_{k+1|k}^R = \mathbf{A}_k \hat{\mathbf{X}}_k^R \quad (39)$$

The error state covariance matrix denoted as  $\mathbf{P}_k$  is updated:

$$\mathbf{P}_{k+1|k} = \mathbf{A}_k \mathbf{P}_k \mathbf{A}_k^T + \mathbf{G}_k \mathbf{Q}_k \mathbf{G}_k^T \quad (40)$$

The state error caused by time delay is defined as  $\delta \mathbf{X}_k$ . The detailed

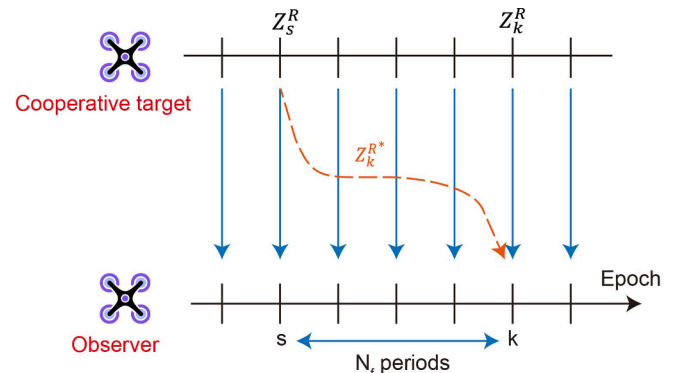


Fig. 7. The delay caused by communication.

derivation process of  $\delta\mathbf{X}_k$  can be seen in the [Supplementary Material](#), and here we directly give its expression:

$$\mathbf{M}_k = \prod_{i=1}^{N_f-1} \mathbf{A}_{k-i}(\mathbf{I} - \mathbf{K}_{k-i}\mathbf{H}_{k-i}) \quad (41)$$

$$\delta\hat{\mathbf{X}}_{k+1|k}^R = \mathbf{M}_k \mathbf{A}_s \mathbf{K}_s (\mathbf{Z}_s^R - \mathbf{H}_s \hat{\mathbf{X}}_{s|s-1}^R) \quad (42)$$

where  $\mathbf{H}_k$  is the observation matrix. In our system, we have three types of observation values: results of DGO, local UWB ranging output, and local camera results. Combining them, we obtain the expression of  $\mathbf{H}_k$ :

$$\mathbf{H}_k = \begin{bmatrix} \mathbf{H}_k^{DGO} \\ \mathbf{H}_k^{UWB} \\ \mathbf{H}_k^{Cam} \end{bmatrix} \quad (43)$$

where  $\mathbf{H}_k^{DGO}$ ,  $\mathbf{H}_k^{UWB}$  and  $\mathbf{H}_k^{Cam}$  are the Jacobian matrixes of corresponding measurement functions:

$$\mathbf{H}_k^{DGO} = \frac{\partial h^{DGO}}{\partial \mathbf{X}}, \mathbf{H}_k^{UWB} = \frac{\partial h^{UWB}}{\partial \mathbf{X}}, \mathbf{H}_k^{Cam} = \frac{\partial h^{Cam}}{\partial \mathbf{X}} \quad (44)$$

The detailed forms of the observation matrix can be found in [Supplementary Material](#). Then, the Kalman gain  $\mathbf{K}_k$  is updated:

$$\mathbf{K}_{k+1} = \mathbf{P}_{k+1|k} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k+1|k} \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \quad (45)$$

The state and covariance matrix can be obtained as

$$\hat{\mathbf{X}}_{k+1|k}^R = \hat{\mathbf{X}}_{k+1|k}^R + \delta\hat{\mathbf{X}}_{k+1|k}^R \quad (46)$$

$$\hat{\mathbf{X}}_{k+1}^R = \hat{\mathbf{X}}_{k+1|k}^R + \mathbf{K}_{k+1} (\mathbf{Z}_s - \mathbf{H}_k \hat{\mathbf{X}}_{s|s-1}^R) \quad (47)$$

$$\mathbf{P}_{k+1} = (\mathbf{I} - \mathbf{K}_{k+1} \mathbf{H}_{k+1}) \mathbf{P}_{k+1|k} \quad (48)$$

The output state  $\hat{\mathbf{X}}_{k+1}^R$  is the final relative positioning result. Compared to the distributed relative position optimization results without filtering, the output state is smoother and has better real-time performance. The algorithm is summarized as follows:

---

**Algorithm 2.** Delay extended Kalman Filter

---

**Input:**

Relative positions optimized by DGO:  $\hat{\mathbf{p}}_{k,(ij)}^R$   
 Local relative measurement results:  $d_{k,(ij)}^{UWB}$ ,  $\theta_{k,(ij)}^{Cam}$   
 Relative velocities:  $\hat{\mathbf{v}}_{k,(ij)}^R$   
 Covariance matrix and noise matrixes:  $\mathbf{P}_k$ ,  $\mathbf{Q}_k$ ,  $\mathbf{R}_k$   
 Period of DEKF and observation values:  $\Delta T_{DEKF}$ ,  $\Delta T_z$

**Output:**

Filtered relative state estimation:  $\hat{\mathbf{X}}_{k+1}^R$

1. Initialize system
  2. **while** (true)
  3. Obtain system time  $t_{sys}$
  4. **if**  $t_{sys} \bmod \Delta T_{DEKF} = 0$
  5. **if**  $t_{sys} \bmod \Delta T_z = 0$
  6. Update  $\hat{\mathbf{X}}_{k+1|k}^R$  and  $\mathbf{P}_{k+1|k}$  with (39), (40)
  7. Estimate state error  $\delta\hat{\mathbf{X}}_{k+1|k}^R$  with (41), (42)
  8. Calculate  $\mathbf{Z}_s^R$
  9. Update  $\hat{\mathbf{X}}_{k+1}^R$  and  $\mathbf{P}_{k+1}$  with (46)-(48)
  10. **else**
  11. Update  $\hat{\mathbf{X}}_{k+1|k}^R$  and  $\mathbf{P}_{k+1|k}$  with (39), (40)
  12. **end if**
  13. **end if**
  14. **end while**
- 

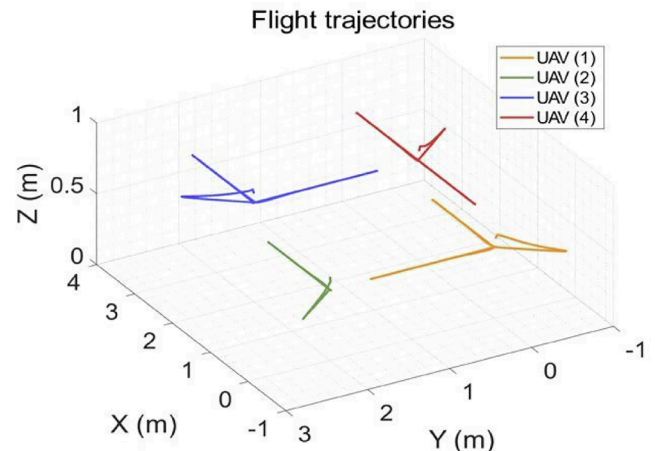
#### 4. Simulation and analysis

To assess the performance of the proposed relative positioning scheme, simulation environment was constructed using MATLAB R2022b. The simulated UAV swarm consists of 4 UAVs numbered 1 ~ 4. The dynamic UAV models provided by MATLAB UAVToolbox were used in our research to effectively simulate real-flight situations of drones. For the onboard sensors, data-driven models were employed in the simulations. Prior to the simulation, we conducted preliminary experiments and data analysis on individual measurement systems alone. The sensor models were then established using the noise, update rate, recognition accuracy and errors extracted from these preliminary experiments to improve the reliability of the simulations. The trajectories of these UAVs are depicted in [Fig. 8](#), which include various flight conditions such as turning, acceleration, deceleration, and other common maneuvers. Initially, the 4 UAVs were placed at (0,0), (0,2), (2,2), and (2,0) (m). The flight process was divided into 4 stages: hovering, expanding, translating, and rotating. These stages allow us to assess the performance of the proposed system under different scenarios. The UAVs moves at a speed of 1.0 m/s. The results were compared with three commonly used methods as a reference. Firstly, we evaluated the original measurement results obtained from onboard optical flow and ToF sensors (denoted as OF&TOF). Secondly, we analyzed the EKF fusion results of the INS which includes an IMU, optical flow sensor, and ToF sensor. Lastly, we examined the DGO results based on EKF fusion results of INS and UWB ranging, without time delay compensation. To guarantee the reliability of the results, The simulation was repeated 50 times, with random measurement noise generated for each iteration.

We adopted the root-mean-square error (RMSE) to evaluate the performances of the relative positioning methods. The RMSE between two UAVs is defined as

$$RMSE = \sqrt{\sum_{m=1}^M \frac{(\hat{x}_m^r - x_m^r)^2 + (\hat{y}_m^r - y_m^r)^2 + (\hat{z}_m^r - z_m^r)^2}{m}} \quad (49)$$

where  $\hat{x}_m^r, \hat{y}_m^r, \hat{z}_m^r$  are the estimated relative coordinates at the  $m$ -th sample point,  $x_m^r, y_m^r, z_m^r$  are the real relative coordinates, and  $m$  is the total number of sample points. In addition to RMSE, other statistics including maximum error, minimum error, and standard deviation (STD) were also recorded. The results are presented in [Fig. 9](#) and [Fig. 10](#). UAV (1) is chosen as a reference, and the relative positioning errors between UAV (1) and UAV (2), (3), (4) are analyzed. The results indicate that the direct measurement method based on optical flow sensor and ToF exhibits the highest error among the four methods due to the cumulative velocity integral error. The EKF method enhances accuracy by fusing the results of multiple sensors of INS; however, the cumulative



**Fig. 8.** The flight trajectories of the UAV swarm in the simulation scenario.

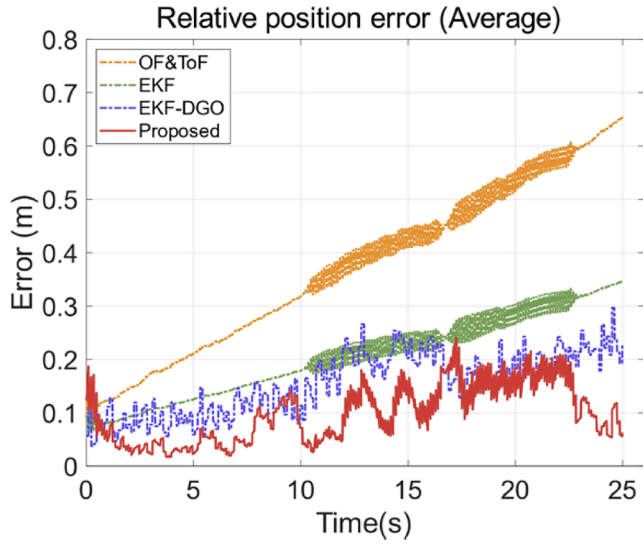


Fig. 9. Error curves of different methods in the simulation.

error persists. The EKF-DGO method aims to reduce INS drift through relative measurements, resulting in a significantly lower RMSE compared to the two navigation-based methods. The proposed method further incorporates additional angle constraint and time delay compensation techniques, exhibiting the highest accuracy. The RMSE, maximum error, minimum error, and STD are consistently smaller than those of the aforementioned methods. Simulations under different

sensor errors were also conducted, and results are shown in the [Supplementary Material](#).

## 5. Real-world experiments

### 5.1. System configuration

To demonstrate the proposed fusion method in real-world experiments, we developed a  $12 \times 12$  cm micro-UAV platform weighing 150 g. The micro-UAV consists of relative measurement, self-state estimation, pre-processing, planning, controlling, and communication modules, as shown in Fig. 11 (a). The types and parameters of the sensors have been introduced in section 2. The YOLOv7-tiny network for target detection runs on a RISC-V AI processor. The fusion, planning, and control algorithms are implemented on two ARMv8 Cortex-M MCUs. Compared to general UAV platforms, the communication frequency and power consumption of the micro-UAV are limited, and we have made a series of optimizations for the engineering deployment of the algorithm. In our system, each UAV communicates with other UAVs through UWB at a 10 Hz rate. The data exchanged between UAVs includes: locally optimized positions of each node, 3-axis velocity of each node, control commands, and other necessary records for data analysis. The total size of all the above-mentioned data is less than 100 bytes. We also conducted a power consumption test, and the total power consumption of sensors and computational units is 2.3 W. Compared to existing micro-UAV systems with computing units of 10 W or more, the proposed system can effectively reduce energy consumption and is suitable for implementation.

The real-world experiment was performed in the university laboratory, as depicted in Fig. 11 (b)(c). A swarm of 4 UAVs marked No.1–4

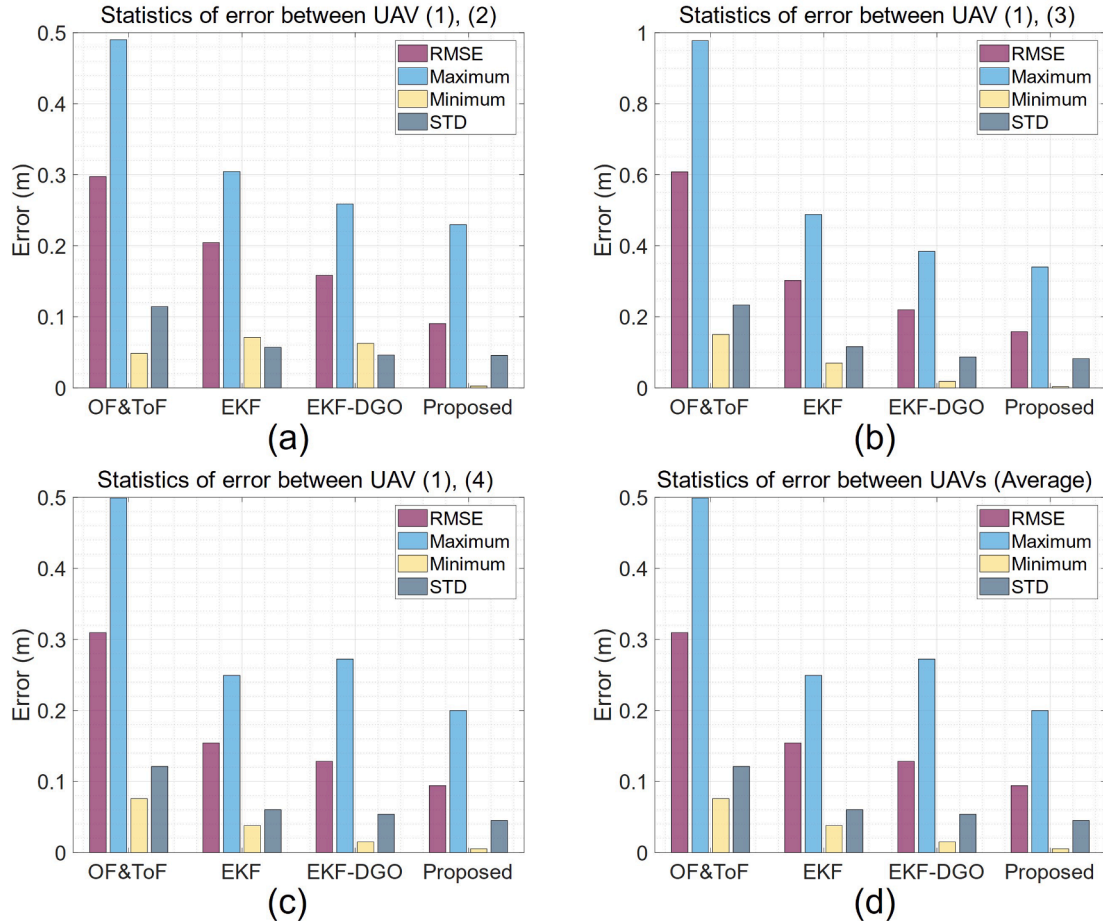
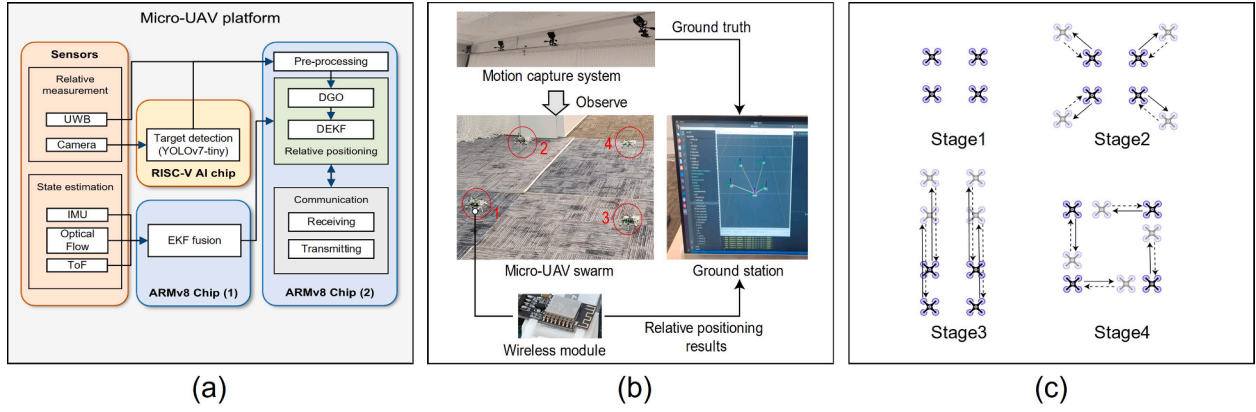


Fig. 10. RMSE, maximum, minimum, and STD of relative positioning error calculated from different methods in the simulation.





**Fig. 11.** Experimental scheme in the lab. (a) System architecture of micro-UAV platform. (b) The photographs of the experimental environment. (c) The pre-set trajectories of the micro-UAV swarm.

flew in the indoor environment and estimated their positions using the onboard algorithm. The results were transmitted to the ground station via a wireless serial port module mounted on UAV (1). The INS results and original sensor outputs were also recorded for comparison and analysis. A NOKOV Mars2HW optical motion capture system acquired the ground truth and sent the true UAV positions to the ground station. The motion capture system was calibrated before the experiment, with a measurement accuracy of  $\pm 0.25$  mm. Note that the motion capture system did not participate in the onboard measurement and controlling processes of UAVs, to ensure the reliability of collected data. As shown in Fig. 11 (c), to verify the feasibility of the algorithm in different flight conditions, the trajectories of the UAVs were pre-set to 4 stages:

Stage 1: the UAVs hovered for 2 s.

Stage 2: the formation shape of UAVs expanded from a small square to a larger one, and then shrunk back.

Stage 3: the UAVs moved to the left side as a whole and flew back.

Stage 4: the formation of UAVs rotated  $90^\circ$  clockwise and transformed back.

Due to the noise of sensors and control errors, the formation cannot strictly maintain geometric shapes as described above. The trajectory error leads to extra challenges to our relative positioning scheme. In such a scenario, the adaptability of the algorithm to the real environment can be well reflected.

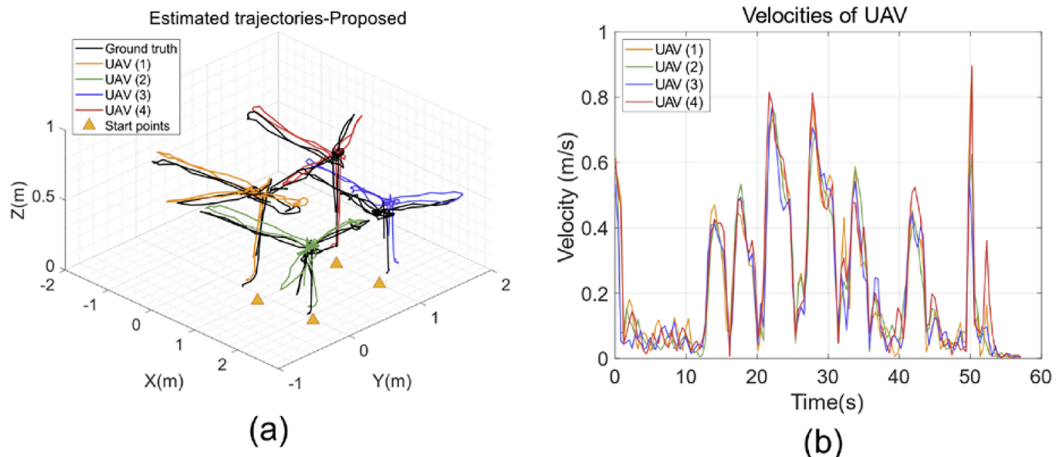
## 5.2. Results and discussion

The performance of the proposed relative positioning system was evaluated by collecting estimated trajectories of the micro-UAV swarm,

as shown in Fig. 12 (a). The estimated trajectories are close to the ground truth in the 3D space. Fig. 12 (b) depicts the UAVs' ground flight speeds, the maximum of which is 0.82 m/s, ignoring the landing process at 50 s. In addition, the maximum relative velocity between UAVs is 0.85 m/s. The algorithm demonstrates a relative positioning accuracy of 0.167 m in real-world UAV systems affected by disturbances and measurement noise, aligning with the simulation results. The error analysis process is detailed below.

For UAV swarm applications, we focus on the relative positioning error between UAVs rather than the absolute positioning error. Taking UAV (1) as a reference, the relative coordinates between UAV (1) and UAV (2), (3), (4) are plotted in Fig. 13. Note that since the relative positioning system runs in a distributed scheme with information interaction, the estimation results of different UAVs are approximately the same. Although the algorithm computes 3D positions of the swarm, the Z-axis coordinates can be directly obtained from the ToF sensor of cm-level accuracy, therefore, the estimated relative X-axis and Y-axis coordinates are displayed in this section. We compare the results of the proposed system with the other three commonly used methods: OF&TOF, EKF, and EKF-DGO method, as mentioned in section 4.

As depicted in Fig. 13, the OF&TOF method has the worst performance both on the X-axis and Y-axis. The cumulative error of the optical flow velocity integral leads to obvious position drift. The EKF method has significantly reduced drift speed by fusing optical flow results with IMU and ToF sensor. However, due to the unpredictable drift direction, the error is amplified in the relative positioning scenario. The EKF-DGO method corrects the cumulative error using UWB ranging results and achieves higher accuracy in the long term. Compared to the above three



**Fig. 12.** The trajectories of the UAV swarm estimated by onboard relative positioning system and the velocities of the UAVs.

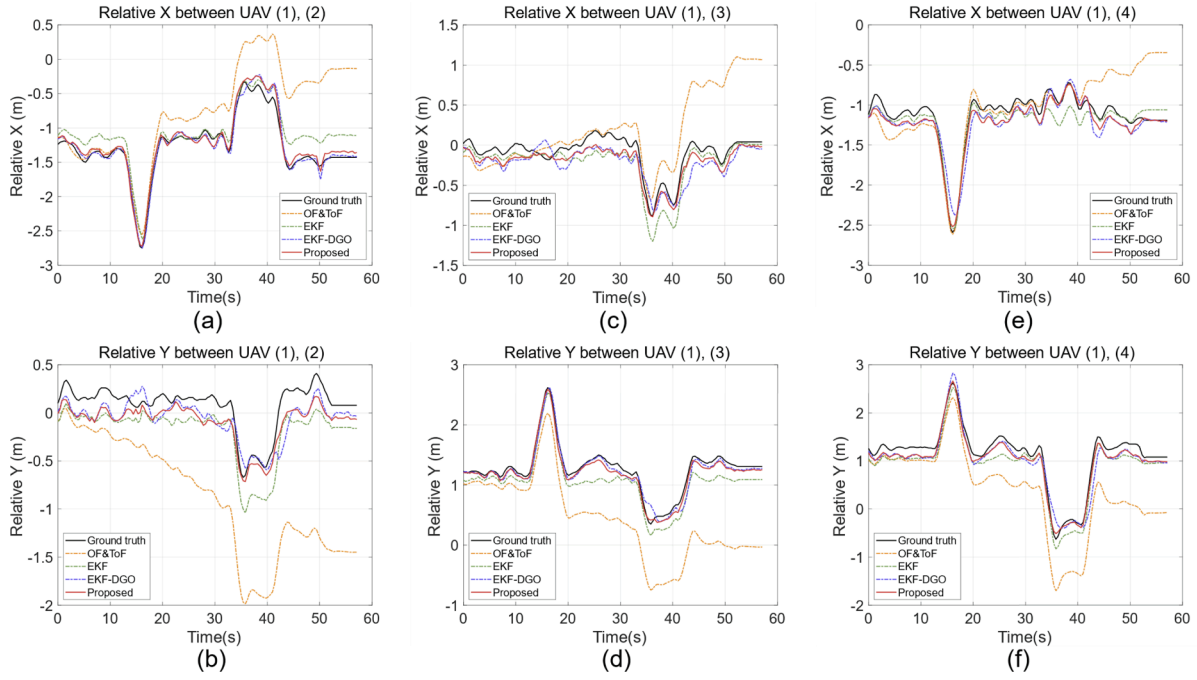


Fig. 13. Estimated relative coordinates between UAV (1) and UAV (2), (3), (4) by different methods.

methods, the proposed relative positioning system has the best performance. The introduction of relative angle constraint and DEKF further improves accuracy and enhances the update rate.

To evaluate the performance of these methods quantitatively, the relative positioning error and cumulative distribution function (CDF) of the methods are shown in Fig. 14. In all three pairs of measurement results, the error of the OF&ToF method exceeds 1 m within 60 s, rendering it unsuitable for swarm tasks. The EKF results of INS exhibit better performance, with errors consistently below 0.5 m. However, without correction of relative measurements, the error continues to increase over time. Predictably, prolonged flights will further deteriorate

the EKF results. On the other hand, the error of EKF-DGO is generally lower than the previous two methods. Nevertheless, the presence of several peaks in the error curve should be noticed. These peaks occur when the relative velocity between UAVs is high enough, indicating that the dynamic property of the EKF-DGO method is insufficient to handle scenes with high relative velocity and limited communication rate. Our proposed method outperforms all three methods. Owing to the contribution of additional relative angular constraints and time delay compensation mechanism, the error was kept at a low level in the flying process. The CDF diagrams also indicate that the proposed method possesses the lowest error on the whole and proves superior to the other

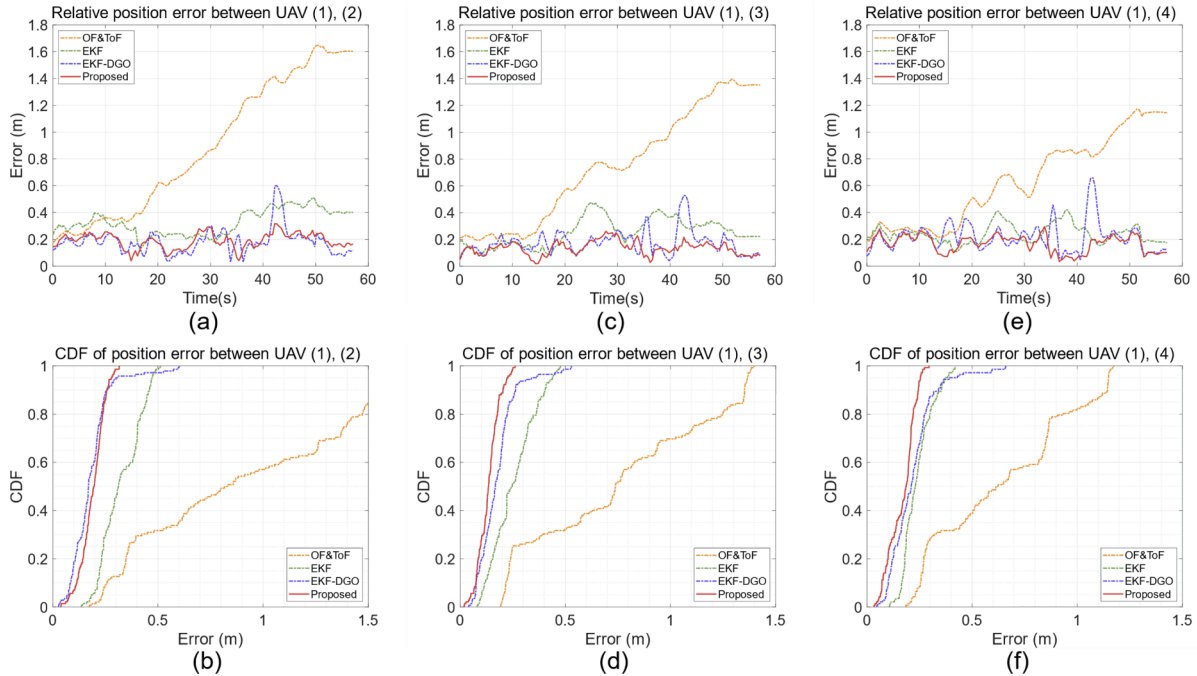


Fig. 14. Relative positioning error and CDF curves.

approaches. Specifically, compared to EKF-DGO, the proposed method demonstrates faster dynamic response and higher update rate.

More specifically, the statistics of relative positioning error including the RMSE, maximum, minimum, and standard deviation (STD) are calculated, as displayed in Fig. 15. The proposed method surpasses others not only in terms of RMSE but also in terms of maximum error and STD. When considering the minimum error, both the proposed method and the EKF-DGO method exhibit similar performance, with a difference of less than 1 cm. Overall, the proposed method provides more accurate, smoother, and stable estimations. The RMSEs of the methods are listed in Table 2. On average, the RMSE of the proposed method is 0.167 m, which represents improvements of 80.7 %, 40.6 %, and 22.3 %, compared to the OF&TOF, EKF, and EKF-DGO methods respectively. This level of relative positioning accuracy will contribute to further cooperative control missions of autonomous micro-UAV swarms.

Furthermore, we investigated the impact of flight speed on positioning accuracy, as illustrated in Fig. 16. Three additional sets of experiments were conducted, where the maximum ground speed ( $V_{max}^{abs}$ ) gradually increased from 0.82 m/s to 2.09 m/s. Simultaneously, the maximum relative speed between UAVs ( $V_{max}^{rel}$ ) was raised from 0.85 m/s to 3.49 m/s. The results indicate that the positioning error grows to a certain extent with the increase in speed, attributed to factors such as higher system latency and greater sensor error under high dynamic conditions. Nevertheless, the proposed method exhibits adaptability to high-speed flights, achieving an average RMSE of 0.193 m at  $V_{max}^{abs} = 2.09$  m/s and  $V_{max}^{rel} = 3.49$  m/s, owing to the delay compensation mechanism. Compared to the OF&ToF, EKF, and EKF-DGO methods, this method shows improvements in RMSE of 84.6 %, 46.4 %, and 24.9 % respectively. The superior dynamic capability of this system makes it

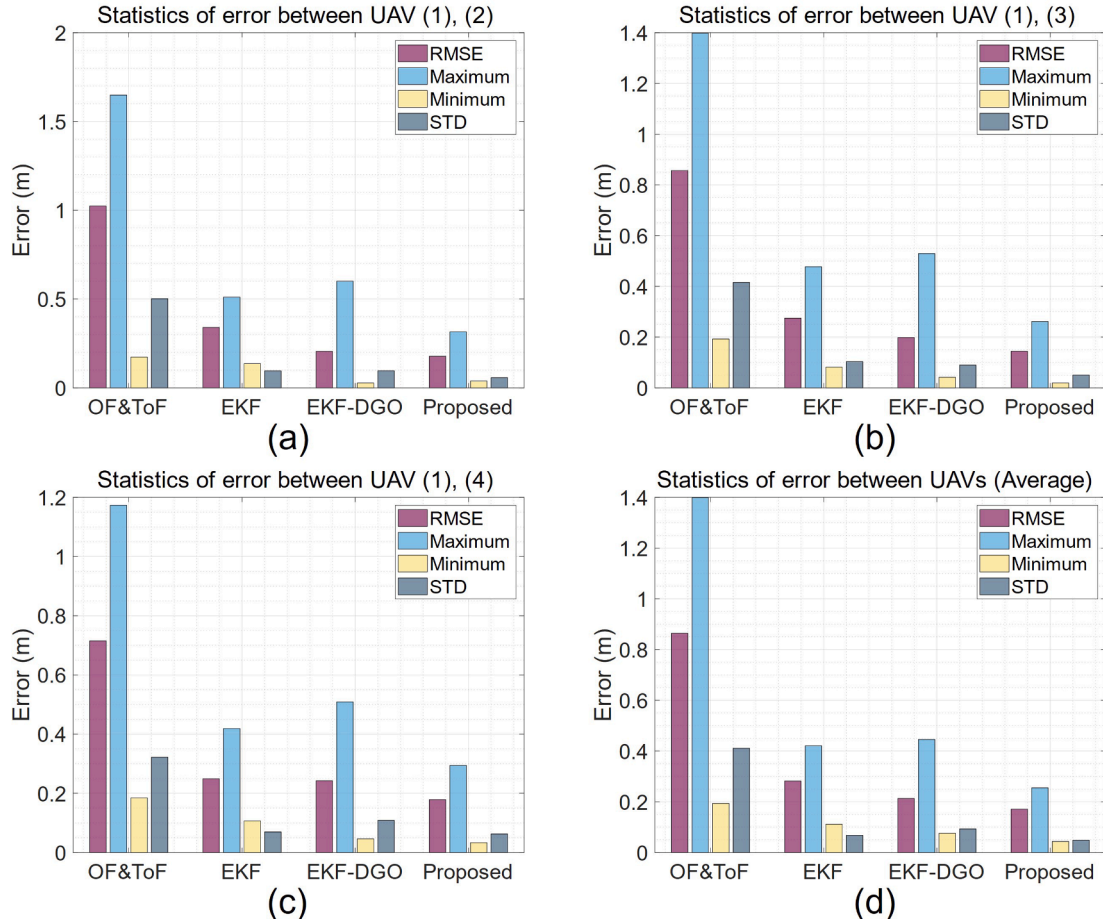
**Table 2**

The RMSEs of the methods.

| Method          | RMSE of relative positioning (m) |              |              |              |
|-----------------|----------------------------------|--------------|--------------|--------------|
|                 | (1), (2)                         | (1), (3)     | (1), (4)     | Average      |
| OF&TOF          | 1.024                            | 0.857        | 0.715        | 0.865        |
| EKF             | 0.340                            | 0.275        | 0.249        | 0.281        |
| EKF-DGO         | 0.205                            | 0.198        | 0.242        | 0.215        |
| <b>Proposed</b> | <b>0.178</b>                     | <b>0.144</b> | <b>0.179</b> | <b>0.167</b> |

suitable for complex UAV swarm missions.

We particularly address the significance of the real-time performance of the proposed distributed optimization scheme on our micro-UAV platform. As comparison, we implemented the proposed DGO architecture and centralized graph optimization (CGO) architecture on the MCU of the UAV system. Measurement data of different node numbers was generated and processed by both architectures. The DGO and CGO were executed for multiple steps, and the average runtime of a single step was calculated, as shown in Fig. 17. When there are only two nodes, the runtime of DGO and CGO is similar. However, as the node number increases to 10, the runtime of CGO exhibits exponential growth, reaching over 80 ms. Since the CGO scheme optimizes the positions of all UAVs, the algorithm faces increased difficulty in converging when various nodes are used. A slight increase in runtime has also been observed for DGO as well, however, DGO still outperforms CGO significantly, with a runtime below 15 ms. The growing trend is caused by the increase in measurement constraints. Due to the fixed dimension of the variable to be optimized, the runtime will not become prohibitively long. In conclusion, the distributed scheme is more suitable for micro-UAVs with limited computing power and demonstrates better scalability. This



**Fig. 15.** RMSE, maximum, minimum, and STD of relative positioning error calculated from different methods in the experiment.

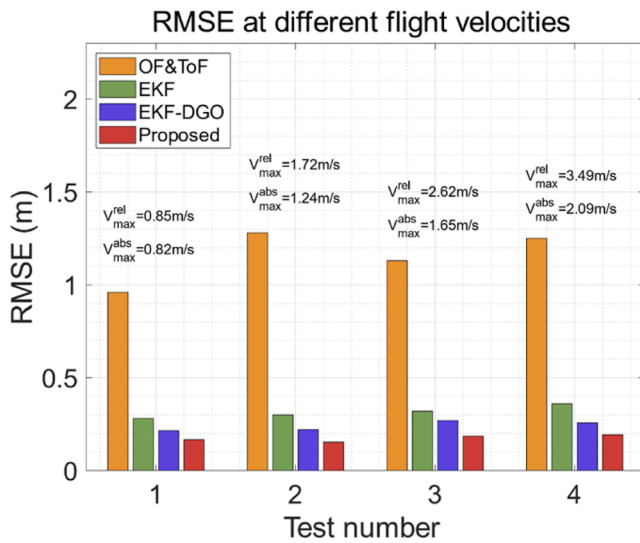


Fig. 16. RMSE at different flight velocities.

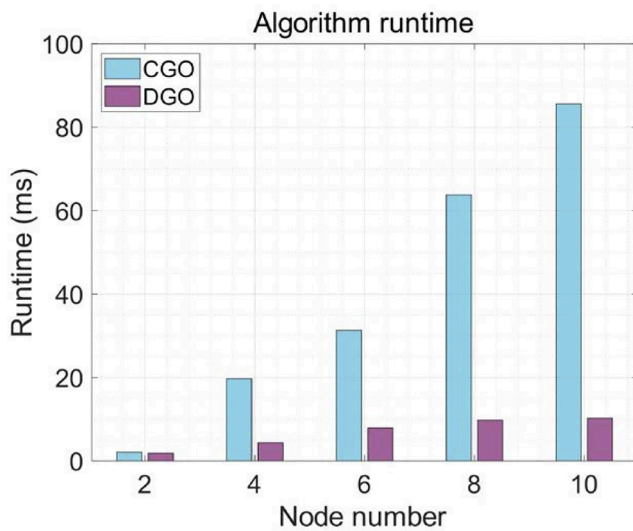


Fig. 17. The runtime of CGO and DGO.

scheme is capable of supporting a swarm comprising various numbers of UAVs.

## 6. Conclusions

In this paper, we have proposed a lightweight robust relative positioning scheme for micro-UAV swarm. This method makes use of DGO to fuse relative distance measurements and a few pairs of relative angle measurements between UAVs, together with state estimations, which all can be achieved by low-cost miniaturized sensors. To further improve positioning accuracy and compensate for error caused by communication delay, a fusion system based on DEKF is designed and applied to the DGO results. Numerical simulation was carried out which indicates the scheme can achieve high-precision relative positioning of UAV swarm without any GNSS, pre-deployed sensor anchors, or wireless base stations. Real-world experiments were conducted using a 4-UAV swarm, and the proposed method achieves a relative positioning accuracy of 0.167 m at a maximum flight velocity of 0.82 m/s, demonstrating an improvement of 22.3 % over the commonly used EKF-DGO method. In high-speed dynamic scenarios with a maximum flight velocity of 2.09 m/s, the proposed method maintains a RMSE of 0.193 m. The

distributed optimization scheme has achieved millisecond-level runtime per step on ARM-Cortex-M level MCU, several times faster than the centralized optimization scheme, suitable for the implementation on actual micro-UAV swarm systems, with the advantages of high-precision and lower hardware requirements.

Our future works include improving the robustness of multi-sensor fusion and exploring relative positioning methods in complex environments where radio communication is jammed.

## CRediT authorship contribution statement

**Chengsong Xiong:** Writing – original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Wenshuai Lu:** Writing – review & editing, Visualization, Validation, Resources, Funding acquisition. **Hao Xiong:** Validation, Software, Methodology. **Haolun Ding:** Validation, Software, Methodology. **Qingning He:** Validation, Software, Methodology. **Duo Zhao:** Validation, Software. **Jiaqi Wan:** Validation, Investigation. **Fei Xing:** Resources, Funding acquisition, Conceptualization. **Zheng You:** Conceptualization, Funding acquisition, Project administration, Resources.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

## Acknowledgement

This research is supported by the National Natural Science Foundation of China (Grant No.U21A6003).

## Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.measurement.2024.114897>.

## References

- [1] Y. Wang, W. Chen, T.H. Luan, Z. Su, Q. Xu, R. Li, N. Chen, Task offloading for post-disaster rescue in unmanned aerial vehicles networks, *IEEE/ACM Transactions on Networking*. 30 (4) (2022) 1525–1539.
- [2] M.M. Nowak, K. Dziób, P. Bogawski, Unmanned Aerial Vehicles (UAVs) in environmental biology: a review, *European Journal of Ecology*. 4 (2) (2019) 56–74.
- [3] P.K. Reddy Maddikunta, S. Hakak, M. Alazab, S. Bhattacharya, T.R. Gadekallu, W. Z. Khan, Q.-V. Pham, Unmanned Aerial Vehicles in Smart Agriculture: Applications, Requirements, and Challenges, *IEEE Sensors Journal*. 21 (16) (2021) 17608–17619.
- [4] R. Kumar, P. Kumar, R. Tripathi, G.P. Gupta, T.R. Gadekallu, G. Srivastava, SP2F: A secured privacy-preserving framework for smart agricultural Unmanned Aerial Vehicles, *Computer Networks*. 187 (2021).
- [5] T.A. Ward, C.J. Fearday, E. Salami, S.N. Binti, A bibliometric review of progress in micro air vehicle research, *International Journal of Micro Air Vehicles*. 9 (2) (2017) 146–165.
- [6] D. Popescu, F. Stoican, G. Stamatescu, O. Chenaru, and L. Ichim, “A Survey of Collaborative UAV-WSN Systems for Efficient Monitoring,” *Sensors (Basel)*, vol. 19, no. 21, Oct 28 2019.
- [7] Lucieer, Arko, et al. “Using a micro-UAV for ultra-high resolution multi-sensor observations of Antarctic moss beds.” (2012): 429.
- [8] A. Bürkle, F. Segor, M. Kollmann, Towards autonomous micro uav swarms, *Journal of Intelligent & Robotic Systems*. 61 (2011) 339–353.
- [9] Y. Guo, M. Wu, K. Tang, J. Tie, X. Li, Covert spoofing algorithm of UAV based on GPS/INS-integrated navigation, *IEEE Transactions on Vehicular Technology*. 68 (7) (2019) 6557–6564.
- [10] Shen S, Mulgaonkar Y, Michael N, Kumar V, editors. Multi-sensor fusion for robust autonomous flight in indoor and outdoor environments with a rotorcraft MAV. 2014 IEEE International Conference on Robotics and Automation (ICRA); 2014: IEEE.



- [11] A. Grant, P. Williams, N. Ward, S. Basker, GPS jamming and the impact on maritime navigation, *The Journal of Navigation*. 62 (2) (2009) 173–187.
- [12] E. Soria, F. Schiano, D. Floreano, Distributed predictive UAV swarms in cluttered environments, *IEEE Robotics and Automation Letters*. 7 (1) (2021) 73–80.
- [13] G. Lu, W. Xu, F. Zhang, On-manifold model predictive control for trajectory tracking on robotic systems, *IEEE Transactions on Industrial Electronics*. 70 (9) (2022) 9192–9202.
- [14] S. Jia, Y. Jiao, Z. Zhang, R. Xiong, Y. Wang, FEJ-VIRO: A Consistent First-Estimate Jacobian Visual-Inertial-Ranging Odometry, in: 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2022, pp. 1336–1343.
- [15] Ledergerber A, Hamer M, D'Andrea R, editors. A robot self-localization system using one-way ultra-wideband communication. 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); 2015: IEEE.
- [16] X. Zhou, X.Y. Wen, Z.P. Wang, Y.M. Gao, H.J. Li, Q.H. Wang, T.K. Yang, H.J. Lu, Y. J. Cao, C. Xu, et al., Swarm of micro flying robots in the wild. *Science, Robotics*. 7 (66) (2022).
- [17] J. Wang, P. Gu, L. Wang, Z.R.V.I.O. Meng, An Effective Localization Algorithm for Range-Aided Visual-Inertial Odometry System, *IEEE Transactions on Intelligent Transportation Systems*. (2023) 1–15.
- [18] H.-Y. Lin, J.-R. Zhan, GNSS-denied UAV indoor navigation with UWB incorporated visual inertial odometry, *Measurement*. 206 (2023).
- [19] A. Fishberg, J.P. How, Multi-Agent Relative Pose Estimation with UWB and Constrained Communications, in: 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, (IROS)2022., pp. 778–785.
- [20] C.C. Cossette, M.A. Shalaby, D. Saussie, J.L. Ny, J.R. Forbes, Optimal Multi-robot Formations for Relative Pose Estimation Using Range Measurements, in: 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems, (IROS) 2022., pp. 2431–2437.
- [21] Y. Jia, J. Li, S. Zhang, Q. Yang, W. Gao, K.S. Kwak, Monocular-Vision-Based Positioning Method for UAV Formation, in: 2022 21st International Symposium on Communications and Information Technologies, (ISCIT)2022., pp. 52–57.
- [22] J. Horyna, V. Walter, M. Saska, UVDAR-COM: UV-Based Relative Localization of UAVs with Integrated Optical Communication, in: 2022 International Conference on Unmanned Aircraft Systems (ICUAS), IEEE, 2022, pp. 1302–1308.
- [23] M. Basiri, F. Schill, P. Lima, D. Floreano, On-Board Relative Bearing Estimation for Teams of UAVs Using Sound, *IEEE Robotics and Automation Letters*. 1 (2) (2016) 820–827.
- [24] Y. Tang, Y. Hu, J. Cui, F. Liao, M. Lao, F. Lin, R.S.H. Teo, Vision-Aided Multi-UAV Autonomous Flocking in GPS-Denied Environment, *IEEE Transactions on Industrial Electronics*. 66 (1) (2019) 616–626.
- [25] Wang C-Y, Bochkovskiy A, Liao H-YM, editors. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition; 2023.
- [26] I. Gallo, A.U. Rehman, R.H. Dehkordi, N. Landro, R. La Grassa, M. Boschetti, Deep object detection of crop weeds: Performance of YOLOv7 on a real case dataset from UAV images, *Remote Sensing*. 15 (2) (2023) 539.
- [27] H. Xu, Y. Zhang, B. Zhou, L. Wang, X. Yao, G. Meng, S. Shen, Omni-Swarm: A Decentralized Omnidirectional Visual-Inertial-UWB State Estimation System for Aerial Swarms, *IEEE Transactions on Robotics*. 38 (6) (2022) 3374–3394.
- [28] Y. Xianjia, L. Qingqing, J.P. Queralta, J. Heikkonen, T. Westerlund, Cooperative UWB-Based Localization for Outdoors Positioning and Navigation of UAVs aided by Ground Robots, in: 2021 IEEE International Conference on Autonomous Systems, (ICAS)2021., pp. 1–5.
- [29] Larsen TD, Andersen NA, Ravn O, Poulsen NK, editors. Incorporation of time delayed measurements in a discrete-time Kalman filter. Proceedings of the 37th IEEE Conference on Decision and Control (Cat. No. 98CH36171); 1998: IEEE.
- [30] A. Fatehi, B. Huang, Kalman filtering approach to multi-rate information fusion in the presence of irregular sampling rate and variable measurement delay, *Journal of Process Control*. 53 (2017) 15–25.
- [31] Liu R, Yuen C, Do T-N, Jiao D, Liu X, Tan U-X, editors. Cooperative relative positioning of mobile users by fusing IMU inertial and UWB ranging information. 2017 IEEE International Conference on Robotics and Automation (ICRA); 2017: IEEE.
- [32] L. Barbieri, M. Brambilla, A. Trabattoni, S. Mervic, M. Nicoli, UWB localization in a smart factory: Augmentation methods and experimental assessment, *IEEE Transactions on Instrumentation and Measurement*. 70 (2021) 1–18.
- [33] M. Cieżkowski, S. Romaniuk, A. Wolniakowski, Apparent beacon position estimation for accuracy improvement in lateration positioning system, *Measurement* 153 (2020), <https://doi.org/10.1016/j.measurement.2019.107400>.
- [34] J. Li, G. Yang, Q. Cai, H. Niu, J. Li, Cooperative navigation for UAVs in GNSS-denied area based on optimized belief propagation, *Measurement*. 192 (2022).
- [35] Q. Fan, B. Sun, Y. Sun, X. Zhuang, Performance Enhancement of MEMS-Based INS/UWB Integration for Indoor Navigation Applications, *IEEE Sensors Journal*. 17 (10) (2017) 3116–3130.
- [36] K. Guo, X. Li, L. Xie, Ultra-wideband and odometry-based cooperative relative localization with application to multi-UAV formation control, *IEEE Transactions on Cybernetics*. 50 (6) (2019) 2590–2603.
- [37] X. Liu, Y. Liu, Y. Chen, Machine Learning Empowered Trajectory and Passive Beamforming Design in UAV-RIS Wireless Networks, *IEEE Journal on Selected Areas in Communications*. 39 (7) (2021) 2042–2055.
- [38] Balamurugan G, Valarmathi J, Naidu V, editors. Survey on UAV navigation in GPS denied environments. 2016 International conference on signal processing, communication, power and embedded system (SCOPES); 2016: IEEE.
- [39] Y. Lu, Z. Xue, G.-S. Xia, L. Zhang, A survey on vision-based UAV navigation, *Geo-Spatial Information Science*. 21 (1) (2018) 21–32.
- [40] Popov A, Miller A, Miller B, Stepanyan K, editors. Optical flow and inertial navigation system fusion in the UAV navigation. Unmanned/unattended sensors and sensor networks XII; 2016: SPIE.
- [41] Chao H, Gu Y, Napolitano M, editors. A survey of optical flow techniques for UAV navigation applications. 2013 International Conference on Unmanned Aircraft Systems (ICUAS); 2013: IEEE.
- [42] Saleem A, Asif KH, Ali A, Awan SM, Alghamdi MA, editors. Pre-processing methods of data mining. 2014 IEEE/ACM 7th International Conference on Utility and Cloud Computing; 2014: IEEE.
- [43] Walter V, Vrba M, Saska M, editors. On training datasets for machine learning-based visual relative localization of micro-scale UAVs. 2020 IEEE International Conference on Robotics and Automation (ICRA); 2020: IEEE.
- [44] S. Minaeian, J. Liu, Y.-J. Son, Vision-based target detection and localization via a team of cooperative UAV and UGVs, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*. 46 (7) (2015) 1005–1016.

**Chengsong Xiong** received his B.S. degree in the Department of Precision Instrument at Tsinghua University in 2020. He currently studies for Ph.D. in Department of Precision Instrument at Tsinghua University. His research interests are in the fields of robotic system and multi-sensor fusion.

**Wenshuai Lu** received his B.S. degree and M.S. degree from Harbin Institute of Technology, and received his Ph.D. degree from Tsinghua University. He is now an assistant research fellow at the Department of Precision Instrument, Tsinghua University. His research interests focus on intelligent integrated microsystems.

**Fei Xing** received the B.S. degree in mechanical engineering degree from Tongji University, Shanghai, China, in 2002, and the Ph.D. degree from Tsinghua University, Beijing, China, in 2006. After graduation, he joined the Faculty of the Department of Precision Instruments and Mechanology, Tsinghua University, as an Assistant Researcher, where he became an Associate Professor in 2011. His main research interests include microminiature high-precision attitude measurement sensor technology. Dr. Xing awarded the specially appointed Professor for Chang Jiang Scholar from the Ministry of Education in 2020.

**Zheng You** received the B.S., M.S., and Ph.D. degrees from the Department of Mechanical Engineering, Huazhong University of Science and Technology, Wuhan, China, in 1985, 1987, and 1990, respectively. He was elected as academicians of Chinese Academy of Engineering in 2013. He is now the president of Huazhong University of Science and Technology, and also a professor at the Department of Precision Instrument, Tsinghua University. His research interests focus on intelligent MEMS and sensors, micro satellites and precision measurement instruments. Dr. You was awarded as Chinese PhD degree holder who has made outstanding Achievements by Commission of Education in 1992, awarded as distinguished professor of Cheung Kong Scholars Program by Ministry of Education in 1999, and awarded as China Excellent post-doctorate by Ministry of Human Resource and Social Security in 2005.